EN.553.761 Unconstrained Nonlinear Optimization

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Contents

0.1 Foundations of Nonlinear Optimization . . . 2
0.1.1 Taylor Formulas and Smoothness . . . 3
0.1.2 Optimality Conditions . . . 4
0.2 Convex Sets and Functions . . . 5
0.2.1 Strong Convexity and Smooth Convexity . . . 6
0.3 Gradient Method . . . 7
0.3.1 Descent Directions and Steepest Descent . . . 7
0.3.2 Descent and Nonconvex Complexity . . . 8
0.3.3 Smooth Convex Objectives . . . 8
0.3.4 Smooth Strongly Convex Objectives . . . 9
0.3.5 Quadratics and Preconditioning . . . 10
0.3.6 Stopping Criteria . . . 10
0.4 Quasi-Newton Method . . . 11
0.4.1 Newton's Method as a Model Method . . . 11
0.4.2 The Secant Equation . . . 11
0.4.3 BFGS, DFP, and SR1 . . . 12
0.4.4 BFGS and Limited-Memory BFGS Algorithms . . . 13
0.4.5 Global and Local Convergence . . . 13
0.5 Line Search . . . 14
0.5.1 Exact Line Search . . . 15
0.5.2 Armijo Backtracking . . . 15
0.5.3 Wolfe Conditions . . . 16
0.5.4 Zoutendijk's Condition . . . 17
0.5.5 Eventual Full Newton Steps . . . 18
0.6 Subgradient Method . . . 18
0.6.1 Subdifferential calculus and optimality . . . 18
0.6.2 Projected subgradient iteration . . . 19
0.6.3 Master bound and step sizes . . . 20
0.6.4 Strongly convex objectives . . . 21
0.7 Proximal Gradient Method . . . 21
0.7.1 Examples of proximal mappings . . . 22
0.7.2 The proximal-gradient step . . . 22
0.7.3 Three-point inequality and descent . . . 23
0.7.4 Convergence rates . . . 23

0.7.5 Backtracking and ISTA . . . 24
0.8 Nesterov's Accelerated Gradient Method . . . 24
0.8.1 Heavy-ball momentum . . . 24
0.8.2 Nesterov acceleration for convex functions . . . 25
0.8.3 Strongly convex acceleration . . . 26
0.8.4 FISTA . . . 26
0.8.5 Restart and safeguards . . . 27
0.9 Frank-Wolfe Method . . . 27
0.9.1 The Frank–Wolfe gap . . . 27
0.9.2 Curvature and descent . . . 28
0.9.3 Primal convergence and gap complexity . . . 28
0.9.4 Linear minimization oracles . . . 29
0.9.5 Away steps and variants . . . 29
0.10 Proximal Point Algorithm . . . 29
0.10.1 Resolvents and the Moreau envelope . . . 30
0.10.2 Inexact proximal steps . . . 31
0.11 Augmented Lagrangian Method . . . 31
0.11.1 ALM as proximal point on the dual . . . 31
0.12 Alternating Direction Method of Multipliers . . . 32
0.12.1 Residuals and stopping criteria . . . 33
0.12.2 Global convergence . . . 33
0.12.3 Two-function splitting and Lasso . . . 34
0.12.4 Consensus and penalty adaptation . . . 34
0.13 Coordinate Descent Method . . . 35
0.14 Stochastic Gradient Descent . . . 35
0.15 Adaptive Methods . . . 36
0.15.1 ADAM . . . 36
0.15.2 AdaGrad . . . 36
0.16 Shampoo and Muon . . . 37
0.17 Automatic Differentiation . . . 37
0.18 Applications . . . 38
0.18.1 Matrix Factorization . . . 38
0.18.2 Support Vector Machines . . . 39
0.18.3 Compressed Sensing . . . 39
0.18.4 Deep Learning . . . 40

0.1 Foundations of Nonlinear Optimization

An unconstrained nonlinear optimization problem has the form

$$\min_{x \in \mathbb{R}^n} f(x),$$

whereas a constrained problem is $\min_{x \in \Omega} f(x)$ for a feasible set $\Omega \subseteq \operatorname{dom} f$. The objective may be smooth or nonsmooth and convex or nonconvex.

Definition 0.1.1 (Local and global solutions)**.** A feasible point x^* is a local solution if some $r > 0$ satisfies

$$f(x^*) \le f(x) \quad \text{for every } x \in \Omega \text{ with } \|x - x^*\| < r.$$

It is a strict local solution if the inequality is strict for every such $x \neq x^*$, and a global solution if it holds for every $x \in \Omega$. The optimal value is $f^* = \inf_{x\in\Omega} f(x)$.

The infimum need not be attained: $\inf_{x\in\mathbb{R}} e^x = 0$, but no finite x achieves it. A continuous function attains its minimum on a nonempty compact set. A useful noncompact alternative is coercivity.

Definition 0.1.2 (Coercivity)**.** A function $f : \mathbb{R}^n \to \mathbb{R}$ is coercive if $f(x) \to +\infty$ whenever $\|x\| \to \infty$.

A lower-semicontinuous coercive function attains its infimum on a nonempty closed feasible set: a sufficiently low sublevel set is bounded, closedness makes it compact, and minimization can be restricted to that compact set.

0.1.1 Taylor Formulas and Smoothness

Differentiability and twice differentiability give the local expansions

$$f(x+p) = f(x) + \nabla f(x)^\mathsf{T} p + o(\|p\|),$$
$$f(x+p) = f(x) + \nabla f(x)^\mathsf{T} p + \frac{1}{2} p^\mathsf{T} \nabla^2 f(x) p + o(\|p\|^2).$$

The integral forms are exact.

Theorem 0.1.1 (Taylor's theorem in integral form)**.** *If f is continuously differentiable on the segment from x to $x+p$, then*

$$f(x+p) = f(x) + \int_0^1 \nabla f(x+tp)^\mathsf{T} p \, \mathrm{d}t. \tag{0.1.1}$$

If f is twice continuously differentiable there, then

$$f(x+p) = f(x) + \nabla f(x)^\mathsf{T} p + \int_0^1 (1-t) p^\mathsf{T} \nabla^2 f(x+tp) p \, \mathrm{d}t. \tag{0.1.2}$$

Proof. Set $\phi(t) = f(x+tp)$. The chain rule gives $\phi'(t) = \nabla f(x+tp)^\mathsf{T} p$, and the fundamental theorem of calculus proves Eq. (0.1.1). Moreover, $\phi''(t) = p^\mathsf{T} \nabla^2 f(x+tp) p$. Applying the fundamental theorem twice and reversing the integration order gives

$$\phi(1) - \phi(0) - \phi'(0) = \int_0^1 \int_0^t \phi''(s) \, \mathrm{d}s \, \mathrm{d}t = \int_0^1 (1-s) \phi''(s) \, \mathrm{d}s,$$

which proves Eq. (0.1.2). □

Definition 0.1.3 (Lipschitz continuity and smoothness)**.** The function f is G-Lipschitz on D if

$$|f(x) - f(y)| \le G \|x - y\|.$$

A differentiable f is L-smooth on D if

$$\|\nabla f(x) - \nabla f(y)\| \le L \|x - y\| .$$

Lemma 0.1.1 (Descent lemma)**.** *If D is convex and f is L-smooth on D, then*

$$f(y) \le f(x) + \nabla f(x)^{\mathsf{T}}(y - x) + \frac{L}{2} \|y - x\|^2 \tag{0.1.3}$$

for all $x, y \in D$.

Proof. Put $p = y - x$ in Eq. (0.1.1). By Cauchy–Schwarz and smoothness,

$$\begin{aligned} f(y) - f(x) - \nabla f(x)^{\mathsf{T}} p &= \int_0^1 \{\nabla f(x + tp) - \nabla f(x)\}^{\mathsf{T}} p \, \mathrm{d}t \\ &\le \int_0^1 Lt \|p\|^2 \, \mathrm{d}t = \frac{L}{2} \|p\|^2 . \end{aligned}$$

□

If f is twice continuously differentiable on an open convex set, the bound $-LI \preceq \nabla^2 f(x) \preceq LI$ implies L-smoothness. Indeed,

$$\nabla f(y) - \nabla f(x) = \int_0^1 \nabla^2 f(x + t(y - x))(y - x) \, \mathrm{d}t.$$

Conversely, L-smoothness implies $\|\nabla^2 f(x)\|_2 \le L$: apply the gradient Lipschitz bound to $x + tv$, divide by $|t|$, and let $t \to 0$ for every unit vector v. For a convex twice differentiable function this becomes $0 \preceq \nabla^2 f(x) \preceq LI$.

0.1.2 Optimality Conditions

Theorem 0.1.2 (Unconstrained first- and second-order conditions)**.** *Let x^* be an interior local minimizer.*

1. *If f is differentiable at x^*, then $\nabla f(x^*) = 0$.*
2. *If f is twice differentiable at x^*, then $\nabla^2 f(x^*) \succeq 0$.*
3. *If $\nabla f(x^*) = 0$, $\nabla^2 f(x^*) \succ 0$, and the Hessian is continuous near x^*, then x^* is a strict local minimizer.*

Proof. If $\nabla f(x^*) \neq 0$, the direction $p = -\nabla f(x^*)$ gives

$$f(x^* + tp) = f(x^*) - t \|\nabla f(x^*)\|^2 + o(t) < f(x^*)$$

for sufficiently small $t > 0$, contradicting local optimality. If some v satisfies $v^{\mathsf{T}} \nabla^2 f(x^*) v < 0$, second-order Taylor expansion along tv produces the same contradiction. Finally, positive definiteness and continuity imply $\nabla^2 f(z) \succeq mI$ in a neighborhood for some $m > 0$; the second-order integral formula then yields $f(x^* + p) \ge f(x^*) + (m/2) \|p\|^2$ locally. □

A point with $\nabla f(x) = 0$ is stationary, but for nonconvex objectives this need not mean local optimality. For example, x^3 is stationary at zero but has no local minimum there. Even a positive-semidefinite Hessian is not sufficient: both x^4 and $-x^4$ have zero Hessian at the origin.

For a nonempty closed convex set Ω, define the normal cone

$$N_\Omega(x) = \{v \in \mathbb{R}^n : v^\mathsf{T}(y - x) \le 0 \text{ for every } y \in \Omega\}.$$

If f is differentiable and convex and Ω is convex, then x^* solves $\min_{x\in\Omega} f(x)$ if and only if

$$0 \in \nabla f(x^*) + N_\Omega(x^*), \tag{0.1.4}$$

equivalently $\nabla f(x^*)^\mathsf{T}(y - x^*) \ge 0$ for every $y \in \Omega$. Necessity follows by differentiating $f(x^* + t(y - x^*))$ at $t = 0$ from the right; sufficiency follows from the convex first-order inequality introduced below.

0.2 Convex Sets and Functions

Definition 0.2.1 (Convex set)**.** A set $C \subseteq \mathbb{R}^n$ is convex if

$$x, y \in C, \quad \theta \in [0, 1] \quad \Longrightarrow \quad \theta x + (1 - \theta)y \in C.$$

Its convex hull is

$$\operatorname{conv}(S) = \left\{ \sum_{i=1}^{m} \theta_i x_i : x_i \in S,\ \theta_i \ge 0,\ \sum_i \theta_i = 1 \right\}.$$

Affine spaces, halfspaces, norm balls, polyhedra, and the positive-semidefinite cone are convex. Arbitrary intersections, affine images, affine inverse images, and coordinate projections preserve convexity. For example, the linear matrix inequality set

$$\left\{ x \in \mathbb{R}^n : A_0 + \sum_{i=1}^{n} x_i A_i \succeq 0 \right\}$$

is convex because it is the affine inverse image of the PSD cone.

If z lies outside a nonempty closed convex set C, a hyperplane strictly separates z from C. To see the geometry, let $x = \operatorname{proj}_C(z)$ and $a = z - x$. Differentiating $\|z - (x + t(y - x))\|^2$ at $t = 0$ gives $a^\mathsf{T}(y - x) \le 0$ for every $y \in C$. Thus with $b = a^\mathsf{T}x$,

$$a^\mathsf{T}y \le b < a^\mathsf{T}z,$$

since $a^\mathsf{T}z - b = \|z - x\|^2 > 0$.

Definition 0.2.2 (Convex function)**.** An extended-real-valued function f is convex if its domain is convex and

$$f(\theta x + (1 - \theta)y) \le \theta f(x) + (1 - \theta)f(y)$$

for all x, y in its domain and $\theta \in [0, 1]$.

Allowing $+\infty$ incorporates constraints: the indicator $\delta_C(x)$ equals 0 on C and $+\infty$ elsewhere, so minimizing $f + \delta_C$ is equivalent to minimizing f over C.

Theorem 0.2.1 (Equivalent convexity tests)**.** *On a convex domain:*

1. *f is convex iff its restriction to every line is convex.*
2. *If f is differentiable, it is convex iff*

$$f(y) \geq f(x) + \nabla f(x)^{\mathsf{T}}(y - x). \tag{0.2.1}$$

3. *If f is twice differentiable on an open domain, it is convex iff $\nabla^2 f(x) \succeq 0$ everywhere.*

Proof. Line restrictions translate the segment definition directly into one dimension. Convexity applied to $x + t(y - x)$, followed by subtraction, division by t, and $t \downarrow 0$, proves Eq. (0.2.1). Conversely, apply that inequality at $z = (1 - \theta)x + \theta y$ separately to x and y; weighting and adding cancels the gradient terms and recovers the segment inequality. Finally, the second derivative of the line restriction is $d^{\mathsf{T}}\nabla^2 f(x + td)d$, so all line restrictions are convex exactly when the Hessian is positive semidefinite. □

The epigraph $\operatorname{epi} f = \{(x, t) : f(x) \leq t\}$ of a convex function is convex, as is every sublevel set. The converse for sublevel sets is false (it characterizes quasiconvexity, not convexity). Convexity is preserved by nonnegative sums, pointwise suprema, affine composition, suitable monotone compositions, perspectives, and partial minimization of jointly convex functions.

0.2.1 Strong Convexity and Smooth Convexity

Definition 0.2.3 (Strong convexity)**.** A differentiable function is μ-strongly convex, $\mu > 0$, if

$$f(y) \geq f(x) + \nabla f(x)^{\mathsf{T}}(y - x) + \frac{\mu}{2}\|y - x\|^2. \tag{0.2.2}$$

Equivalently, $f(x) - (\mu/2)\|x\|^2$ is convex. For twice differentiable f, this is equivalent to $\nabla^2 f(x) \succeq \mu I$ everywhere.

Strong convexity is stronger than strict convexity: x^4 is strictly convex but not strongly convex on $\mathbb{R}$. If f is differentiable and strongly convex on $\mathbb{R}^n$, it has a unique minimizer x^* and

$$f(x) - f^* \geq \frac{\mu}{2}\|x - x^*\|^2, \tag{0.2.3}$$

$$f(x) - f^* \leq \frac{1}{2\mu}\|\nabla f(x)\|^2, \tag{0.2.4}$$

$$\|\nabla f(x)\| \geq \mu\|x - x^*\|. \tag{0.2.5}$$

The first follows from Eq. (0.2.2) at x^*. For the second, minimize the quadratic lower model in y, whose minimizer is $y = x - \nabla f(x)/\mu$. The third follows from strong monotonicity,

$$\{\nabla f(x) - \nabla f(y)\}^{\mathsf{T}}(x - y) \geq \mu\|x - y\|^2,$$

with $y = x^*$ and Cauchy–Schwarz.

For convex L-smooth functions, the gradient is $1/L$-cocoercive:

$$\{\nabla f(x) - \nabla f(y)\}^\mathsf{T}(x - y) \geq \frac{1}{L} \|\nabla f(x) - \nabla f(y)\|^2 . \tag{0.2.6}$$

To derive it, first note that any convex L-smooth h minimized at u satisfies, by applying the descent lemma at $v - \nabla h(v)/L$,

$$h(v) - h(u) \geq \frac{1}{2L} \|\nabla h(v)\|^2 .$$

Apply this to $h_x(z) = f(z) - \nabla f(x)^\mathsf{T} z$, which is minimized at x, once with (x, y) and once with their roles reversed. Adding the resulting inequalities proves Eq. (0.2.6).

If f and Ω are convex, every local minimizer over Ω is global: otherwise, the segment toward a better feasible point would contain arbitrarily close points with smaller objective. The set of global minimizers is convex, and strict convexity permits at most one. Smoothness therefore provides a global quadratic upper model, strong convexity a quadratic lower model, and the condition number $\kappa = L/\mu$ will govern first-order rates.

0.3 Gradient Method

Throughout this section, f is continuously differentiable and bounded below by $f^* > -\infty$. Convexity is imposed only when stated.

0.3.1 Descent Directions and Steepest Descent

A vector d is a descent direction at x if some $\bar{\alpha} > 0$ satisfies $f(x + \alpha d) < f(x)$ for every $\alpha \in (0, \bar{\alpha}]$. If $\nabla f(x)^\mathsf{T} d < 0$, then d is a descent direction: continuity makes $\nabla f(x + td)^\mathsf{T} d < 0$ for all sufficiently small $t \geq 0$, and

$$f(x + \alpha d) - f(x) = \int_0^\alpha \nabla f(x + td)^\mathsf{T} d \, \mathrm{d}t < 0.$$

In Euclidean geometry, the unit direction of steepest local decrease solves

$$\min_{\|d\| \leq 1} \nabla f(x)^\mathsf{T} d = - \|\nabla f(x)\| , \qquad d = -\frac{\nabla f(x)}{\|\nabla f(x)\|}.$$

More generally, for a norm $\|\cdot\|$ with dual norm $\|\cdot\|_*$, the minimum equals $-\|\nabla f(x)\|_*$. Thus "steepest" depends on geometry; preconditioned and Newton directions amount to choosing a different local norm.

0.3.2 Descent and Nonconvex Complexity

Gradient descent uses

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k). \tag{0.3.1}$$

For an L-smooth objective and constant step α, the descent lemma gives

$$f(x - \alpha \nabla f(x)) \le f(x) - \alpha \left(1 - \frac{L\alpha}{2}\right) \|\nabla f(x)\|^2 . \tag{0.3.2}$$

Every $\alpha \in (0, 2/L)$ therefore produces strict descent at a nonstationary point. The choice $\alpha = 1/L$ maximizes the guaranteed decrease:

$$f(x_{k+1}) \le f(x_k) - \frac{1}{2L} \|\nabla f(x_k)\|^2 .$$

Theorem 0.3.1 (Global stationarity complexity)**.** *If f is L-smooth and bounded below, gradient descent with constant $\alpha \in (0, 2/L)$ satisfies*

$$\min_{0 \le k < T} \|\nabla f(x_k)\|^2 \le \frac{f(x_0) - f^*}{T\alpha(1 - L\alpha/2)}. \tag{0.3.3}$$

For $\alpha = 1/L$,

$$\min_{0 \le k < T} \|\nabla f(x_k)\| \le \sqrt{\frac{2L\{f(x_0) - f^*\}}{T}}.$$

Thus $O(L(f(x_0) - f^*)/\varepsilon^2)$ iterations suffice to find an ε-stationary point.

Proof. Sum Eq. (0.3.2) from $k = 0$ to $T - 1$ and use $f(x_T) \ge f^*$. The minimum of the T squared gradient norms is no larger than their average. □

The same argument shows $\sum_{k=0}^{\infty} \|\nabla f(x_k)\|^2 < \infty$ and therefore $\|\nabla f(x_k)\| \to 0$. It does not by itself show that the full iterate sequence converges or that any limit is a local minimum.

0.3.3 Smooth Convex Objectives

Theorem 0.3.2 (Sublinear convergence)**.** *Suppose f is convex and L-smooth with minimizer x^*. For any constant $\alpha \in (0, 1/L]$, gradient descent satisfies*

$$f(x_k) - f^* \le \frac{\|x_0 - x^*\|^2}{2\alpha k}, \qquad k \ge 1. \tag{0.3.4}$$

For $\alpha = 1/L$, this becomes

$$f(x_k) - f(x^*) \le \frac{L \|x_0 - x^*\|^2}{2k}.$$

Proof. Write $g_k = \nabla f(x_k)$. Descent and convexity yield

$$\begin{aligned} f(x_{k+1}) &\le f(x_k) - \frac{\alpha}{2}\|g_k\|^2 \\ &\le f^* + g_k^{\mathsf{T}}(x_k - x^*) - \frac{\alpha}{2}\|g_k\|^2 \\ &= f^* + \frac{1}{2\alpha}\left(\|x_k - x^*\|^2 - \|x_{k+1} - x^*\|^2\right). \end{aligned}$$

Summing telescopes. Since the objective values are nonincreasing, the final gap is no larger than the average of the first k gaps, proving Eq. (0.3.4). □

Smooth convexity also gives the useful sandwich

$$\frac{1}{2L}\|\nabla f(x)\|^2 \le f(x) - f^* \le \frac{L}{2}\|x - x^*\|^2. \tag{0.3.5}$$

The left inequality follows because f^* is no larger than $f(x - \nabla f(x)/L)$ and the descent lemma bounds that trial value. The right follows by applying the descent lemma with base point x^* and using $\nabla f(x^*) = 0$.

0.3.4 Smooth Strongly Convex Objectives

Theorem 0.3.3 (Linear convergence in function value)**.** *If f is also μ-strongly convex, gradient descent with $\alpha = 1/L$ satisfies*

$$f(x_k) - f(x^*) \le \left(1 - \frac{\mu}{L}\right)^k \{f(x_0) - f(x^*)\}.$$

Thus $O(\kappa \log((f(x_0) - f^*)/\varepsilon))$ iterations suffice for a function gap at most ε, where $\kappa = L/\mu$.

Proof. Strong convexity implies the Polyak–Łojasiewicz inequality $\|\nabla f(x)\|^2 \ge 2\mu(f(x) - f^*)$ by Eq. (0.2.4). Substituting it into the $1/L$ descent bound gives

$$f(x_{k+1}) - f^* \le \left(1 - \frac{\mu}{L}\right)(f(x_k) - f^*),$$

and iteration proves the claim. □

A slightly different step is optimal for worst-case contraction in distance. For an L-smooth, μ-strongly convex function,

$$\{\nabla f(x) - \nabla f(y)\}^{\mathsf{T}}(x - y) \ge \frac{\mu L}{\mu + L}\|x - y\|^2 + \frac{1}{\mu + L}\|\nabla f(x) - \nabla f(y)\|^2. \tag{0.3.6}$$

Expanding one gradient step, using Eq. (0.3.6) with $y = x^*$, and choosing $\alpha = 2/(L + \mu)$ cancels the gradient-square term and gives

$$\|x_{k+1} - x^*\| \le \frac{L - \mu}{L + \mu}\|x_k - x^*\| = \frac{\kappa - 1}{\kappa + 1}\|x_k - x^*\|. \tag{0.3.7}$$

0.3.5 Quadratics and Preconditioning

For

$$f(x) = \frac{1}{2}x^{\mathsf{T}}Qx - b^{\mathsf{T}}x, qquadQ \succ 0,$$

the minimizer is $x^* = Q^{-1}b$ and the error obeys

$$e_{k+1} = (I - \alpha Q)e_k, qquade_k = x_k - x^*.$$

If the eigenvalues of Q lie in $[\mu, L]$, the worst contraction factor is $\max_{\lambda\in[\mu,L]} |1-\alpha\lambda|$. Balancing the two endpoints again gives $\alpha = 2/(L+\mu)$ and factor $(\kappa-1)/(\kappa+1)$.

Exact line search along the negative gradient does at least as well as that trial step. In the energy norm $\|e\|_Q^2 = e^{\mathsf{T}}Qe$,

$$\|(I - 2Q/(L+\mu))e\|_Q \le \frac{L-\mu}{L+\mu}\|e\|_Q .$$

Since $f(x) - f^* = \frac{1}{2}\|x - x^*\|_Q^2$, steepest descent with exact line search satisfies

$$f(x_{k+1}) - f^* \le \left(\frac{\kappa-1}{\kappa+1}\right)^2 (f(x_k) - f^*).$$

Large κ produces elongated contours and zig-zagging. A positive-definite preconditioner changes the update to

$$x_{k+1} = x_k - \alpha P\nabla f(x_k),$$

which is steepest descent in the norm $\|d\|_{P^{-1}} = (d^{\mathsf{T}}P^{-1}d)^{1/2}$. For a quadratic, the ideal but usually impractical choice $P = Q^{-1}$ solves the problem in one step with $\alpha = 1$. Quasi-Newton methods try to learn this inverse curvature.

0.3.6 Stopping Criteria

For a general nonconvex objective, gradient norm is the primary first-order stopping measure. Strong convexity additionally provides the computable certificates

$$f(x_k) - f^* \le \frac{\|\nabla f(x_k)\|^2}{2\mu}, \qquad \|x_k - x^*\| \le \frac{\|\nabla f(x_k)\|}{\mu}.$$

In floating-point computation, use a scaled test such as

$$\|\nabla f(x_k)\|_\infty \le \varepsilon_{\text{abs}} + \varepsilon_{\text{rel}} \max\{1, |f(x_k)|\},$$

together with safeguards based on step norm and an iteration budget. Objective change alone can be misleading on flat regions.

0.4 Quasi-Newton Method

0.4.1 Newton's Method as a Model Method

At x_k, write $g_k = \nabla f(x_k)$ and form the quadratic model

$$m_k(p) = f(x_k) + g_k^{\mathsf{T}} p + \frac{1}{2} p^{\mathsf{T}} \nabla^2 f(x_k) p.$$

If $\nabla^2 f(x_k) \succ 0$, its minimizer is the Newton direction, computed by solving

$$\nabla^2 f(x_k) p_k^N = -g_k \tag{0.4.1}$$

rather than forming an inverse. It is a descent direction because

$$g_k^{\mathsf{T}} p_k^N = -g_k^{\mathsf{T}} \nabla^2 f(x_k)^{-1} g_k < 0$$

whenever $g_k \neq 0$. Far from a solution, the Hessian may be indefinite or the quadratic model inaccurate, so a globalized method uses $x_{k+1} = x_k + \alpha_k p_k^N$ with line search or a trust region.

Theorem 0.4.1 (Local quadratic convergence of Newton's method)**.** *Suppose $\nabla f(x^*) = 0$, $\nabla^2 f(x^*) \succ 0$, the Hessian is locally M-Lipschitz, and $\nabla^2 f(x) \succeq \mu I$ in a neighborhood of x^*. For x_k sufficiently close to x^*, the unit Newton step satisfies*

$$\|x_{k+1} - x^*\| \leq \frac{M}{2\mu} \|x_k - x^*\|^2 . \tag{0.4.2}$$

Proof. Let $e_k = x_k - x^*$ and $G_k = \nabla^2 f(x_k)$. The fundamental theorem of calculus and $\nabla f(x^*) = 0$ give

$$\nabla f(x_k) = \int_0^1 \nabla^2 f(x^* + t e_k) e_k \, \mathrm{d}t.$$

Therefore

$$e_{k+1} = G_k^{-1} \int_0^1 \{G_k - \nabla^2 f(x^* + t e_k)\} e_k \, \mathrm{d}t.$$

Since $\left\|G_k^{-1}\right\|_2 \leq 1/\mu$ and Hessian Lipschitzness bounds the integrand by $M(1-t)\|e_k\|^2$, integration proves Eq. (0.4.2). □

0.4.2 The Secant Equation

Quasi-Newton methods replace the Hessian or inverse Hessian with an estimate. Set $s_k = x_{k+1} - x_k$ and $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$. The secant The fundamental theorem of calculus gives

$$y_k = \left\{ \int_0^1 \nabla^2 f(x_k + t s_k) \, \mathrm{d}t \right\} s_k, \tag{0.4.3}$$

so y_k is the action of an average Hessian on s_k. This motivates the secant conditions

$$B_{k+1} s_k = y_k, \qquad H_{k+1} y_k = s_k,$$

for Hessian and inverse-Hessian approximations. One vector equation cannot determine a symmetric $n \times n$ matrix, so quasi-Newton updates additionally change the previous approximation as little as possible in a suitable weighted matrix norm.

0.4.3 BFGS, DFP, and SR1

The BFGS update is

$$B_{k+1} = B_k - \frac{B_k s_k s_k^{\mathsf{T}} B_k}{s_k^{\mathsf{T}} B_k s_k} + \frac{y_k y_k^{\mathsf{T}}}{y_k^{\mathsf{T}} s_k}. \tag{0.4.4}$$

In inverse form, with $\rho_k = (y_k^{\mathsf{T}} s_k)^{-1}$,

$$H_{k+1} = (I - \rho_k s_k y_k^{\mathsf{T}}) H_k (I - \rho_k y_k s_k^{\mathsf{T}}) + \rho_k s_k s_k^{\mathsf{T}}, \tag{0.4.5}$$

while the DFP inverse update is

$$H_{k+1} = H_k - \frac{H_k y_k y_k^{\mathsf{T}} H_k}{y_k^{\mathsf{T}} H_k y_k} + \frac{s_k s_k^{\mathsf{T}}}{y_k^{\mathsf{T}} s_k}.$$

BFGS and DFP are dual members of the Broyden family.

The symmetric-rank-one update is

$$B_{k+1} = B_k + \frac{(y_k - B_k s_k)(y_k - B_k s_k)^{\mathsf{T}}}{(y_k - B_k s_k)^{\mathsf{T}} s_k}. \tag{0.4.6}$$

SR1 can represent indefinite curvature, which is useful in nonconvex trust-region methods. It is skipped when its denominator is too small relative to $\|y_k - B_k s_k\| \, \|s_k\|$.

All these formulas satisfy their respective secant equations. For example, multiplying Eq. (0.4.4) by s_k gives

$$B_k s_k - B_k s_k \frac{s_k^{\mathsf{T}} B_k s_k}{s_k^{\mathsf{T}} B_k s_k} + y_k \frac{y_k^{\mathsf{T}} s_k}{y_k^{\mathsf{T}} s_k} = y_k.$$

For SR1, writing $u_k = y_k - B_k s_k$ yields $(B_k + u_k u_k^{\mathsf{T}}/(u_k^{\mathsf{T}} s_k)) s_k = y_k$.

Theorem 0.4.2 (BFGS preserves positive definiteness)**.** *If $H_k \succ 0$ and $y_k^{\mathsf{T}} s_k > 0$, then the inverse BFGS update satisfies $H_{k+1} \succ 0$. Consequently $p_k = -H_k g_k$ is a descent direction whenever $g_k \neq 0$.*

Proof. Let $V = I - \rho_k y_k s_k^{\mathsf{T}}$. For every nonzero z,

$$z^{\mathsf{T}} H_{k+1} z = (Vz)^{\mathsf{T}} H_k (Vz) + \rho_k (s_k^{\mathsf{T}} z)^2.$$

Both terms are nonnegative. If both vanished, then $Vz = 0$ and $s_k^{\mathsf{T}} z = 0$, which together imply $z = 0$, a contradiction. □

If f is μ-strongly convex, Eq. (0.4.3) gives $y_k^{\mathsf{T}} s_k \geq \mu \|s_k\|^2 > 0$. More generally, a Wolfe line search guarantees the curvature condition for a descent direction.

0.4.4 BFGS and Limited-Memory BFGS Algorithms

Full BFGS computes $p_k = -H_k g_k$, chooses a Wolfe step α_k, sets $s_k = \alpha_k p_k$, evaluates $y_k = g_{k+1} - g_k$, and applies Eq. (0.4.5). It requires $O(n^2)$ storage and matrix-vector work per iteration. The common initial choice is $H_0 = I$; after a curvature pair is available, the scalar initialization

$$H_k^0 = \gamma_k I, \qquad \gamma_k = \frac{s_{k-1}^\mathsf{T} y_{k-1}}{y_{k-1}^\mathsf{T} y_{k-1}}$$

matches inverse curvature along the most recent direction.

L-BFGS stores only the most recent m pairs, often $5 \le m \le 20$, and applies the implicit inverse by the two-loop recursion.

Algorithm 1 L-BFGS two-loop recursion for $p_k = -H_k g_k$

$q \leftarrow g_k$ and $\rho_i \leftarrow 1/(y_i^\mathsf{T} s_i)$.
for $i = k-1, k-2, \ldots, k-m$ **do**
    $a_i \leftarrow \rho_i s_i^\mathsf{T} q$; $q \leftarrow q - a_i y_i$.
end for
$r \leftarrow H_k^0 q$.
for $i = k-m, k-m+1, \ldots, k-1$ **do**
    $b_i \leftarrow \rho_i y_i^\mathsf{T} r$; $r \leftarrow r + s_i(a_i - b_i)$.
end for
return $p_k = -r$.

The time and storage cost is $O(mn)$; use only the pairs that exist when $k < m$. Fixed-memory L-BFGS is scalable but does not generally inherit full BFGS's local superlinear convergence theorem.

0.4.5 Global and Local Convergence

Under a representative global result, suppose f is twice continuously differentiable and convex, the initial level set is bounded, and $0 < \mu I \preceq \nabla^2 f(x) \preceq LI$ near that set. BFGS initialized with $H_0 \succ 0$ and using a Wolfe line search satisfies $\nabla f(x_k) \to 0$ and $x_k \to x^*$.

The logic combines three ingredients. Curvature bounds give

$$\mu \|s_k\|^2 \le y_k^\mathsf{T} s_k, \qquad \|y_k\| \le L \|s_k\|,$$

so the updates remain positive definite. The Wolfe search and Zoutendijk theory control the gradient projected along the search directions. Finally, the BFGS identities

$$\operatorname{tr}(B_{k+1}) = \operatorname{tr}(B_k) - \frac{\|B_k s_k\|^2}{s_k^\mathsf{T} B_k s_k} + \frac{\|y_k\|^2}{y_k^\mathsf{T} s_k},$$

$$\det(B_{k+1}) = \det(B_k) \frac{y_k^\mathsf{T} s_k}{s_k^\mathsf{T} B_k s_k}$$

prevent the search directions from becoming nearly orthogonal to the negative gradient too persistently. Merely writing down the BFGS update does not itself establish global convergence.

For the local result, let $B^* = \nabla^2 f(x^*)$. The Dennis–Moré condition is

$$\frac{\|(B_k - B^*)p_k\|}{\|p_k\|} \longrightarrow 0. \tag{0.4.7}$$

It requires curvature accuracy along the steps, not entrywise convergence of B_k.

Theorem 0.4.3 (Local BFGS behavior)**.** *For a twice continuously differentiable, strongly convex objective with Lipschitz Hessian near x^*, BFGS with a suitable Wolfe line search converges superlinearly once unit steps are accepted and Eq. (0.4.7) holds:*

$$\frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} \longrightarrow 0.$$

Proof. Let $e_k = x_k - x^*$. Hessian Lipschitzness gives $\nabla f(x_k) = B^* e_k + O(\|e_k\|^2)$. For an eventual unit step, $e_{k+1} = e_k + p_k$ and $B_k p_k = -\nabla f(x_k)$, so

$$B^* e_{k+1} = B^* e_k - \nabla f(x_k) + (B^* - B_k)p_k.$$

Uniform positive definiteness and local gradient Lipschitzness imply $\|p_k\| = O(\|e_k\|)$. The first term on the right is $O(\|e_k\|^2)$, while Eq. (0.4.7) makes the second $o(\|p_k\|) = o(\|e_k\|)$. Since B^* is nonsingular, $\|e_{k+1}\| = o(\|e_k\|)$. □

In computation, solve Newton systems rather than invert Hessians. Regularize an indefinite Hessian using $H + \lambda I$, use a trust region, or switch to a descent direction.

0.5 Line Search

Given a descent direction p_k with $g_k^{\mathsf{T}} p_k < 0$, line search chooses $\alpha_k > 0$ and sets $x_{k+1} = x_k + \alpha_k p_k$. Exact line search minimizes $\phi(\alpha) = f(x_k + \alpha p_k)$ but is usually unnecessary.

Merely requiring $f(x_{k+1}) < f(x_k)$ is too weak: vanishing steps can converge to a nonstationary point. A useful analysis instead seeks a uniform reduction

$$f(x_{k+1}) \le f(x_k) - C\,\|g_k\|^2, \qquad C > 0. \tag{0.5.1}$$

If f is bounded below, summing this inequality gives

$$C \sum_{k=0}^{T-1} \|g_k\|^2 \le f(x_0) - f(x_T) \le f(x_0) - f^*.$$

Hence $\sum_k \|g_k\|^2 < \infty$, $g_k \to 0$, and

$$\min_{0 \le k < T} \|g_k\| \le \sqrt{\frac{f(x_0) - f^*}{CT}}.$$

For convex f, every accumulation point is globally optimal. If f is μ-strongly convex and $2\mu C < 1$, the PL inequality yields

$$f(x_{k+1}) - f^* \le (1 - 2\mu C)\{f(x_k) - f^*\}.$$

0.5.1 Exact Line Search

Exact line search can be expensive and is rarely warranted when the direction itself comes from a local approximation. For a quadratic

$$f(x) = \frac{1}{2}x^\mathsf{T}Qx - b^\mathsf{T}x, \qquad Q \succ 0,$$

and steepest-descent direction $p_k = -g_k$, however, it has the closed form

$$\alpha_k = \frac{g_k^\mathsf{T} g_k}{g_k^\mathsf{T} Q g_k}. \tag{0.5.2}$$

Indeed,

$$\phi(\alpha) = f(x_k) - \alpha g_k^\mathsf{T} g_k + \frac{\alpha^2}{2} g_k^\mathsf{T} Q g_k,$$

and solving $\phi'(\alpha) = 0$ proves Eq. (0.5.2). Since $g_{k+1} = g_k - \alpha_k Q g_k$, consecutive gradients satisfy $g_{k+1}^\mathsf{T} g_k = 0$. This orthogonality explains steepest descent's zig-zagging on poorly conditioned quadratics.

0.5.2 Armijo Backtracking

Definition 0.5.1 (Armijo condition)**.** The Armijo sufficient-decrease condition is

$$f(x_k + \alpha p_k) \le f(x_k) + c_1 \alpha g_k^\mathsf{T} p_k, \qquad 0 < c_1 < 1.$$

Backtracking starts from an initial step, often one, and repeatedly multiplies it by $\beta \in (0, 1)$ until Armijo holds.

Algorithm 2 Backtracking line search

1: Choose $\alpha \leftarrow \bar{\alpha}$, $\beta \in (0, 1)$, $c_1 \in (0, 1)$.
2: **while** $f(x + \alpha p) > f(x) + c_1 \alpha \nabla f(x)^\mathsf{T} p$ **do**
3: $\quad \alpha \leftarrow \beta \alpha$.
4: **end while**
5: Return α.

Theorem 0.5.1 (Finite termination and a step lower bound)**.** *Suppose ∇f is L-Lipschitz on the search segment and $g^\mathsf{T} p < 0$. Armijo holds whenever*

$$0 < \alpha \le \frac{2(1 - c_1)(-g^\mathsf{T} p)}{L \|p\|^2}. \tag{0.5.3}$$

Consequently, backtracking terminates finitely and accepts

$$\alpha \geq \min\left\{\bar{\alpha}, \frac{2\beta(1-c_1)(-g^{\mathsf{T}}p)}{L\|p\|^2}\right\}. \tag{0.5.4}$$

Proof. The descent lemma gives

$$f(x+\alpha p) \leq f(x) + \alpha g^{\mathsf{T}}p + \frac{L\alpha^2}{2}\|p\|^2.$$

This upper bound implies Armijo when $(1-c_1)\alpha g^{\mathsf{T}}p + (L\alpha^2/2)\|p\|^2 \leq 0$, which is Eq. (0.5.3). The geometric trial sequence must enter this interval. If the initial trial is too large, the last rejected trial exceeds the right side of Eq. (0.5.3); multiplying it by β gives Eq. (0.5.4). □

For gradient descent, $p = -g$, so the accepted step is uniformly bounded below by

$$\underline{\alpha} = \min\left\{\bar{\alpha}, \frac{2\beta(1-c_1)}{L}\right\} > 0,$$

and

$$f(x_{k+1}) \leq f(x_k) - c_1\underline{\alpha}\|g_k\|^2.$$

Thus the stationarity and strongly convex conclusions following Eq. (0.5.1) apply.

0.5.3 Wolfe Conditions

Write $\phi(\alpha) = f(x_k + \alpha p_k)$. The Wolfe conditions are

$$\phi(\alpha) \leq \phi(0) + c_1\alpha\phi'(0), \tag{0.5.5}$$
$$\phi'(\alpha) \geq c_2\phi'(0), \qquad 0 < c_1 < c_2 < 1. \tag{0.5.6}$$

Since $\phi'(0) < 0$, the curvature condition requires the slope to become substantially less negative, preventing an unnecessarily short step. The strong Wolfe condition

$$|\phi'(\alpha)| \leq c_2|\phi'(0)|$$

also prevents stepping too far beyond a one-dimensional minimizer.

Theorem 0.5.2 (Existence of Wolfe steps)**.** *Suppose ϕ is continuously differentiable, bounded below for $\alpha > 0$, and $\phi'(0) < 0$. For every $0 < c_1 < c_2 < 1$, some positive interval contains steps satisfying both Wolfe conditions.*

Proof. Define $\psi(\alpha) = \phi(\alpha) - \phi(0) - c_1\alpha\phi'(0)$. Then $\psi(0) = 0$ and $\psi'(0) = (1-c_1)\phi'(0) < 0$, so $\psi < 0$ initially. Because ϕ is bounded below while $\phi(0) + c_1\alpha\phi'(0) \to -\infty$, eventually $\psi > 0$. Let $\bar{\alpha}$ be the first subsequent zero; Armijo holds on $(0, \bar{\alpha}]$. If curvature failed throughout this interval, then $\phi'(\alpha) < c_2\phi'(0)$ there. Integration would give

$$\phi(\bar{\alpha}) - \phi(0) < c_2\bar{\alpha}\phi'(0) < c_1\bar{\alpha}\phi'(0),$$

contradicting $\psi(\bar{\alpha}) = 0$. □

For BFGS, $s_k = \alpha_k p_k$ and $y_k = g_{k+1} - g_k$. Wolfe curvature implies

$$\begin{aligned} y_k^\mathsf{T} s_k &= \alpha_k (g_{k+1}^\mathsf{T} p_k - g_k^\mathsf{T} p_k) \\ &\geq \alpha_k (c_2 - 1) g_k^\mathsf{T} p_k > 0, \end{aligned}$$

so the BFGS inverse approximation remains positive definite. Armijo alone does not guarantee this on a nonconvex objective.

0.5.4 Zoutendijk's Condition

Define the angle between $-g_k$ and p_k by

$$\cos\theta_k = -\frac{g_k^\mathsf{T} p_k}{\|g_k\| \, \|p_k\|}.$$

A direction sequence is gradient related when, for fixed positive constants,

$$\cos\theta_k \geq \epsilon_0, \qquad \gamma_1 \|g_k\| \leq \|p_k\| \leq \gamma_2 \|g_k\| .$$

Theorem 0.5.3 (Zoutendijk condition)**.** *Assume f is bounded below on its initial level set, its gradient is Lipschitz on a neighborhood of that set, every p_k is a descent direction, and every step satisfies Wolfe. Then*

$$\sum_{k=0}^{\infty} \cos^2\theta_k \, \|g_k\|^2 < \infty. \tag{0.5.7}$$

If $\cos\theta_k \geq \epsilon_0 > 0$, then $g_k \to 0$.

Proof. Gradient Lipschitzness along the step and Wolfe curvature imply

$$\alpha_k \geq \frac{(1 - c_2)(-g_k^\mathsf{T} p_k)}{L \, \|p_k\|^2}.$$

Substitute this into Armijo:

$$f(x_k) - f(x_{k+1}) \geq \frac{c_1(1 - c_2)}{L} \frac{(g_k^\mathsf{T} p_k)^2}{\|p_k\|^2} = \frac{c_1(1 - c_2)}{L} \cos^2\theta_k \, \|g_k\|^2 .$$

Summing telescopes and f is bounded below, proving Eq. (0.5.7). A uniform acute-angle bound then makes the squared gradient norms summable. □

The theorem separates responsibilities: the line search guarantees progress along the chosen direction, while the direction-generation method must avoid directions becoming nearly orthogonal to $-g_k$. Newton directions are gradient related when $\mu I \preceq \nabla^2 f(x) \preceq LI$, since

$$\frac{-g^\mathsf{T} p}{\|g\| \, \|p\|} \geq \frac{\mu}{L}, \qquad \frac{1}{L} \|g\| \leq \|p\| \leq \frac{1}{\mu} \|g\| .$$

Similar bounds hold for $p_k = -H_k g_k$ when H_k is uniformly positive definite and bounded.

0.5.5 Eventual Full Newton Steps

Theorem 0.5.4 (Eventual acceptance of the unit Newton step)**.** *Suppose the Hessian is Lipschitz near x^*, $\nabla f(x^*) = 0$, and $\nabla^2 f(x^*) \succ 0$. If $c_1 < 1/2$, then sufficiently near x^* the Newton step satisfies Armijo with $\alpha = 1$. Backtracking Newton therefore eventually recovers quadratic convergence.*

Proof. Let $g = \nabla f(x)$, $H = \nabla^2 f(x)$, and $p = -H^{-1}g$. Hessian Lipschitzness with constant M gives

$$f(x+p) \le f(x) + g^\mathsf{T} p + \frac{1}{2} p^\mathsf{T} H p + \frac{M}{6}\|p\|^3 = f(x) + \frac{1}{2} g^\mathsf{T} p + \frac{M}{6}\|p\|^3 .$$

Near x^*, $H \succeq \mu I$, so $-g^\mathsf{T} p = p^\mathsf{T} H p \ge \mu \|p\|^2$. The unit step satisfies Armijo whenever

$$\frac{M}{6}\|p\| \le \left(\frac{1}{2} - c_1\right)\mu.$$

As $x \to x^*$, $p \to 0$, so this eventually holds. □

For BFGS, Wolfe search both preserves positive definiteness and, under standard local assumptions, eventually accepts unit steps, enabling superlinear convergence.

0.6 Subgradient Method

Let $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be proper and convex. A vector g is a *subgradient* at $x \in \operatorname{dom} f$ if

$$f(y) \ge f(x) + g^\mathsf{T}(y - x) \qquad \text{for every } y \in \mathbb{R}^n.$$

The closed convex set of all such vectors is the *subdifferential* $\partial f(x)$. If f is differentiable at x, then $\partial f(x) = \{\nabla f(x)\}$; in general it may contain many supporting slopes. Important examples are

$$\partial |x| = \begin{cases} \{1\}, & x > 0, \\ [-1,1], & x = 0, \\ \{-1\}, & x < 0, \end{cases} \qquad \partial \|x\|_2 = \begin{cases} \{x/\|x\|_2\}, & x \ne 0, \\ \{g : \|g\|_2 \le 1\}, & x = 0. \end{cases}$$

More generally, for a norm and its dual norm,

$$\partial \|x\| = \{g : \|g\|_* \le 1,\ g^\mathsf{T} x = \|x\|\}.$$

0.6.1 Subdifferential calculus and optimality

For proper convex functions, with the usual relative-interior qualification where needed,

$$\begin{aligned} \partial(af)(x) &= a\partial f(x), && a > 0, \\ \partial(f_1 + \cdots + f_m)(x) &= \partial f_1(x) + \cdots + \partial f_m(x), \\ \partial\Big(f(Ax+b)\Big) &= A^\mathsf{T} \partial f(Ax+b). \end{aligned}$$

If $f(x) = \max_i f_i(x)$ and $I(x) = \{i : f_i(x) = f(x)\}$, then

$$\partial f(x) = \operatorname{conv}\left(\bigcup_{i \in I(x)} \partial f_i(x) \right).$$

Equivalently, the one-sided directional derivative is the support function of the subdifferential:

$$f'(x; d) = \max_{g \in \partial f(x)} g^{\mathsf{T}} d.$$

Thus an arbitrary negative subgradient need not be a descent direction. The minimum-norm subgradient $g_{\min}$ is special: away from a minimizer, $f'(x; -g_{\min}) = -\|g_{\min}\|^2 < 0$.

The nonsmooth Fermat rule is

$$x^* \in \arg\min_x f(x) \quad \Longleftrightarrow \quad 0 \in \partial f(x^*).$$

For minimization over a nonempty closed convex set C, define

$$N_C(x) = \{v : v^{\mathsf{T}}(y - x) \le 0 \text{ for every } y \in C\}.$$

Since $N_C(x) = \partial \delta_C(x)$, constrained optimality becomes

$$0 \in \partial f(x^*) + N_C(x^*).$$

For example, $0 \in \nabla h(x^*) + \lambda \partial \|x^*\|_1$ explains the coordinatewise thresholding behavior of ℓ_1 regularization.

0.6.2 Projected subgradient iteration

Choose $s_k \in \partial f(x_k)$ and update

$$x_{k+1} = \operatorname{proj}_C(x_k - \alpha_k s_k),$$

where $C = \mathbb{R}^n$ gives the unconstrained method. Euclidean projection is nonexpansive. Indeed, if $p = \operatorname{proj}_C(u)$ and $q = \operatorname{proj}_C(v)$, projection optimality gives

$$(u - p)^{\mathsf{T}}(q - p) \le 0, \qquad (v - q)^{\mathsf{T}}(p - q) \le 0.$$

Adding these inequalities and applying Cauchy–Schwarz yields

$$\|p - q\|^2 \le (u - v)^{\mathsf{T}}(p - q) \le \|u - v\| \, \|p - q\|,$$

hence $\|p - q\| \le \|u - v\|$.

Let x^* minimize f over C. Nonexpansiveness and the subgradient inequality give the fundamental distance recursion

$$\begin{aligned} \|x_{k+1} - x^*\|^2 &\le \|x_k - \alpha_k s_k - x^*\|^2 \\ &= \|x_k - x^*\|^2 - 2\alpha_k s_k^{\mathsf{T}}(x_k - x^*) + \alpha_k^2 \|s_k\|^2 \\ &\le \|x_k - x^*\|^2 - 2\alpha_k \Big(f(x_k) - f^* \Big) + \alpha_k^2 \|s_k\|^2. \end{aligned}$$

This distance-based analysis is necessary because the objective values need not decrease from one iteration to the next.

0.6.3 Master bound and step sizes

Assume $\|x_0 - x^*\| \le R$ and $\|s_k\| \le G$. Summing the distance recursion for $k = 0, \dots, T-1$ gives

$$\sum_{k=0}^{T-1} \alpha_k \Big(f(x_k) - f^* \Big) \le \frac{R^2 + G^2 \sum_{k=0}^{T-1} \alpha_k^2}{2}.$$

For the weighted average

$$\bar{x}_T = \frac{\sum_{k=0}^{T-1} \alpha_k x_k}{\sum_{k=0}^{T-1} \alpha_k},$$

convexity, and separately the fact that a minimum is no larger than an average, produce the common master bound

$$\max\Big\{ f(\bar{x}_T) - f^*,\ \min_{0 \le k < T} f(x_k) - f^* \Big\} \le \frac{R^2 + G^2 \sum_{k=0}^{T-1} \alpha_k^2}{2 \sum_{k=0}^{T-1} \alpha_k}. \tag{0.6.1}$$

With a constant step $\alpha_k = \alpha$,

$$f(\bar{x}_T) - f^* \le \frac{R^2}{2T\alpha} + \frac{G^2 \alpha}{2}.$$

For a known horizon, $\alpha = R/(G\sqrt{T})$ balances the terms and gives

$$f(\bar{x}_T) - f^* \le \frac{RG}{\sqrt{T}},$$

so $O(R^2 G^2 / \epsilon^2)$ iterations suffice for accuracy ϵ. If $\alpha_k \to 0$ and $\sum_k \alpha_k = \infty$, then

$$\frac{\sum_{k<T} \alpha_k^2}{\sum_{k<T} \alpha_k} \longrightarrow 0,$$

so the best-iterate gap vanishes. The stronger classical conditions $\sum_k \alpha_k = \infty$ and $\sum_k \alpha_k^2 < \infty$ also control the distance sequence.

When f^* is known, Polyak's step

$$\alpha_k = \frac{f(x_k) - f^*}{\|s_k\|^2}$$

turns the recursion into

$$\|x_{k+1} - x^*\|^2 \le \|x_k - x^*\|^2 - \frac{\big(f(x_k) - f^*\big)^2}{\|s_k\|^2}.$$

Thus distance to every solution decreases monotonically and the best gap is again at most $RG/\sqrt{T}$. An overestimate of f^* destroys this guarantee.

0.6.4 Strongly convex objectives

If f is μ-strongly convex, then

$$s_k^{\mathsf{T}}(x_k - x^*) \geq f(x_k) - f^* + \frac{\mu}{2}\|x_k - x^*\|^2,$$

and therefore

$$\|x_{k+1} - x^*\|^2 \leq (1 - \mu\alpha_k)\|x_k - x^*\|^2 - 2\alpha_k\Big(f(x_k) - f^*\Big) + \alpha_k^2 G^2.$$

Taking $\alpha_k = 1/(\mu k)$ for $k = 1, \ldots, T$ and rearranging gives

$$f(x_k) - f^* \leq \frac{\mu}{2}\Big((k-1)\|x_k - x^*\|^2 - k\|x_{k+1} - x^*\|^2\Big) + \frac{G^2}{2\mu k}.$$

The distance terms telescope. For $\bar{x}_T = T^{-1}\sum_{k=1}^{T} x_k$, convexity then yields

$$f(\bar{x}_T) - f^* \leq \frac{G^2(1 + \log T)}{2\mu T}.$$

More careful weighting removes the logarithm and gives $O(G^2/(\mu T))$. Despite its broad applicability, the basic method can be nonmonotone and step-size sensitive; proximal, bundle, smoothing, or primal–dual methods exploit additional structure more effectively.

0.7 Proximal Gradient Method

Consider the composite convex problem

$$\min_x F(x) = f(x) + h(x),$$

where f is convex and differentiable with L-Lipschitz gradient, while h is proper, closed, and convex but may be nonsmooth or extended-valued. This includes constraints through $h = \delta_C$ and sparsity penalties through $h(x) = \lambda\|x\|_1$.

Definition 0.7.1 (Proximal operator)**.** For $\alpha > 0$,

$$\operatorname{prox}_{\alpha h}(v) = \arg\min_x \left\{h(x) + \frac{1}{2\alpha}\|x - v\|^2\right\}.$$

The objective defining the proximal map is $1/\alpha$-strongly convex, so its minimizer exists and is unique. Its optimality condition is

$$p = \operatorname{prox}_{\alpha h}(v) \quad \Longleftrightarrow \quad \frac{v - p}{\alpha} \in \partial h(p). \tag{0.7.1}$$

The proximal map is *firmly nonexpansive*. If $p = \operatorname{prox}_{\alpha h}(u)$ and $q = \operatorname{prox}_{\alpha h}(v)$, monotonicity of ∂h gives

$$0 \leq \left\langle p - q, \frac{u - p}{\alpha} - \frac{v - q}{\alpha}\right\rangle,$$
$$\|p - q\|^2 \leq \langle p - q, u - v\rangle.$$

Cauchy–Schwarz therefore implies ordinary nonexpansiveness.

0.7.1 Examples of proximal mappings

If $h(x) = \sum_i h_i(x_i)$, the proximal problem separates coordinatewise. For $\tau = \alpha\lambda$, this gives soft thresholding:

$$\Big[\operatorname{prox}_{\alpha\lambda\|\cdot\|_1}(v)\Big]_i = \operatorname{sign}(v_i)(|v_i| - \tau)_+.$$

Indeed, the scalar condition $0 \in \lambda\partial\,|p| + (p - v)/\alpha$ yields $p = v - \tau$ for $v > \tau$, $p = v + \tau$ for $v < -\tau$, and $p = 0$ for $|v| \le \tau$. Other useful formulas include

$$\begin{aligned}
\operatorname{prox}_{\alpha\lambda\|\cdot\|_2}(v) &= \left(1 - \frac{\tau}{\|v\|_2}\right)_+ v, \\
\operatorname{prox}_{\alpha(\lambda/2)\|\cdot\|_2^2}(v) &= \frac{v}{1 + \alpha\lambda}, \\
\operatorname{prox}_{\alpha\delta_C}(v) &= \operatorname{proj}_C(v).
\end{aligned}$$

For a matrix $V = U\operatorname{diag}(\sigma_i)W^{\mathsf{T}}$, singular-value thresholding gives

$$\operatorname{prox}_{\alpha\lambda\|\cdot\|_*}(V) = U\operatorname{diag}\Big((\sigma_i - \tau)_+\Big)W^{\mathsf{T}}.$$

Thus coordinate sparsity, group sparsity, low rank, and simple constraints can all be handled by choosing an appropriate proximal map.

The Moreau decomposition is sometimes the easiest route to a formula. With h^* the convex conjugate,

$$v = \operatorname{prox}_{\alpha h}(v) + \alpha\operatorname{prox}_{h^*/\alpha}(v/\alpha).$$

For example, since $(\lambda\|\cdot\|_\infty)^*$ is the indicator of the ℓ_1 ball of radius λ,

$$\operatorname{prox}_{\alpha\lambda\|\cdot\|_\infty}(v) = v - \operatorname{proj}_{\{z:\|z\|_1 \le \alpha\lambda\}}(v).$$

0.7.2 The proximal-gradient step

The descent lemma motivates the quadratic model

$$Q_\alpha(y; x) = f(x) + \langle\nabla f(x), y - x\rangle + \frac{1}{2\alpha}\|y - x\|^2 + h(y).$$

When $0 < \alpha \le 1/L$, this model majorizes $F(y)$. Completing the square shows that its minimizer is the proximal-gradient step

$$x_{k+1} = \operatorname{prox}_{\alpha h}\Big(x_k - \alpha\nabla f(x_k)\Big). \tag{0.7.2}$$

The associated gradient mapping

$$G_\alpha(x) = \frac{1}{\alpha}\left[x - \operatorname{prox}_{\alpha h}(x - \alpha\nabla f(x))\right]$$

reduces to $\nabla f(x)$ when $h = 0$. By (0.7.1),

$$G_\alpha(x) = 0 \quad\Longleftrightarrow\quad 0 \in \nabla f(x) + \partial h(x) \quad\Longleftrightarrow\quad x \in \arg\min F.$$

0.7.3 Three-point inequality and descent

Let $x^+ = \operatorname{prox}_{\alpha h}(x - \alpha \nabla f(x))$. Proximal optimality supplies

$$r^+ = \frac{x - x^+}{\alpha} - \nabla f(x) \in \partial h(x^+).$$

For any $y \in \operatorname{dom} h$, combine the subgradient inequality for h, convexity of f, and the descent lemma:

$$\begin{aligned} F(x^+) - F(y) &\le \langle \nabla f(x), x^+ - y \rangle + \frac{L}{2} \left\| x^+ - x \right\|^2 + \langle r^+, x^+ - y \rangle \\ &= \frac{1}{\alpha} \langle x - x^+, x^+ - y \rangle + \frac{L}{2} \left\| x^+ - x \right\|^2. \end{aligned}$$

Using

$$2\langle x - x^+, x^+ - y \rangle = \|x - y\|^2 - \left\| x^+ - y \right\|^2 - \left\| x - x^+ \right\|^2$$

gives the central three-point estimate

$$F(x^+) \le F(y) + \frac{1}{2\alpha} \left(\|x - y\|^2 - \left\| x^+ - y \right\|^2 - (1 - \alpha L) \left\| x^+ - x \right\|^2 \right). \tag{0.7.3}$$

A slightly sharper comparison with $y = x$ in the strongly convex proximal model gives, for $0 < \alpha < 2/L$,

$$F(x^+) \le F(x) - \alpha \left(1 - \frac{\alpha L}{2} \right) \|G_\alpha(x)\|^2. \tag{0.7.4}$$

Hence the usual choice $\alpha \le 1/L$ decreases the objective.

0.7.4 Convergence rates

Theorem 0.7.1 (Proximal-gradient rate)**.** *If F is convex, f is L-smooth, and a minimizer exists, then with $\alpha = 1/L$,*

$$F(x_k) - F(x^*) \le \frac{L \|x_0 - x^*\|^2}{2k}.$$

To prove the theorem, set $y = x^*$ and $\alpha = 1/L$ in (0.7.3):

$$F(x_{j+1}) - F^* \le \frac{L}{2} \left(\|x_j - x^*\|^2 - \|x_{j+1} - x^*\|^2 \right).$$

Summing from $j = 0$ to $k - 1$ telescopes. Since the objective values are nonincreasing by (0.7.4),

$$k\Big(F(x_k) - F^*\Big) \le \sum_{j=1}^{k} \Big(F(x_j) - F^*\Big) \le \frac{L}{2} \|x_0 - x^*\|^2.$$

If f is additionally μ-strongly convex, the same proof gains the term $-(\mu/2)\|x_k - x^*\|^2$. Combining it with strong convexity of F yields

$$(L + \mu) \|x_{k+1} - x^*\|^2 \le (L - \mu) \|x_k - x^*\|^2,$$

and consequently

$$\|x_k - x^*\|^2 \le \left(\frac{L-\mu}{L+\mu}\right)^k \|x_0 - x^*\|^2 .$$

Even when f is nonconvex, (0.7.4) remains valid for convex h. If F is bounded below and $0 < \alpha < 2/L$, summing it gives

$$\min_{0\le j<k} \|G_\alpha(x_j)\|^2 \le \frac{F(x_0) - \inf F}{k\alpha(1 - \alpha L/2)},$$

so an ϵ-stationary point requires $O(1/\epsilon^2)$ iterations.

0.7.5 Backtracking and ISTA

When L is unknown, repeatedly replace a trial step by $\alpha \leftarrow \beta\alpha$, where $\beta \in (0,1)$, until the smooth-term majorization test

$$f(x^+) \le f(x) + \langle \nabla f(x), x^+ - x\rangle + \frac{1}{2\alpha}\left\|x^+ - x\right\|^2$$

holds. This test terminates once $\alpha \le 1/L$; it must be applied to the smooth function f, not the full objective F.

For the Lasso,

$$\min_x \frac{1}{2}\|Ax - b\|^2 + \lambda \|x\|_1 ,$$

proximal gradient becomes the iterative shrinkage-thresholding algorithm (ISTA):

$$x_{k+1} = \mathrm{soft}_{\alpha_k\lambda}\Big(x_k - \alpha_k A^\mathsf{T}(Ax_k - b)\Big).$$

Its accelerated version, FISTA, appears in the next section.

0.8 Nesterov's Accelerated Gradient Method

Momentum methods retain information from two consecutive iterates. They can reduce the zig-zagging caused by ill-conditioning, but their guarantees depend strongly on how and where the gradient is evaluated.

0.8.1 Heavy-ball momentum

Polyak's heavy-ball iteration is

$$x_{k+1} = x_k - \alpha\nabla f(x_k) + \beta(x_k - x_{k-1}), \qquad \alpha > 0, \quad 0 \le \beta < 1. \tag{0.8.1}$$

Equivalently, $v_{k+1} = \beta v_k + \nabla f(x_k)$ and $x_{k+1} = x_k - \alpha v_{k+1}$, so the velocity averages recent gradients. For the quadratic

$$f(x) = \frac{1}{2}x^\mathsf{T}Qx - b^\mathsf{T}x, qquadQ \succ 0,$$

the error in an eigendirection of Q with eigenvalue λ obeys

$$e_{k+1}^{(\lambda)} = (1 + \beta - \alpha\lambda)e_k^{(\lambda)} - \beta e_{k-1}^{(\lambda)}.$$

Thus stability is determined by the roots of $r^2 - (1 + \beta - \alpha\lambda)r + \beta = 0$ for every $\lambda \in [\mu, L]$.

For positive-definite quadratics, the parameters

$$\alpha_{\mathrm{HB}} = \frac{4}{(\sqrt{L} + \sqrt{\mu})^2}, \qquad \beta_{\mathrm{HB}} = \left(\frac{\sqrt{L} - \sqrt{\mu}}{\sqrt{L} + \sqrt{\mu}}\right)^2$$

give spectral radius at most

$$\rho_{\mathrm{HB}} = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}, \qquad \kappa = \frac{L}{\mu}.$$

To see this, write $q = (\sqrt{L} - \sqrt{\mu})/(\sqrt{L} + \sqrt{\mu})$. The middle coefficient of the scalar recurrence ranges from $2q$ to $-2q$, while the root product is q^2. The endpoint roots are repeated roots $\pm q$, and interior complex-conjugate roots have modulus q. These parameters are exact for quadratics but need not be stable for a general smooth convex function.

0.8.2 Nesterov acceleration for convex functions

Nesterov's method evaluates the gradient at an extrapolated point. Set $x_0 = y_1$ and $t_1 = 1$, and for $k \geq 1$ compute

$$x_k = y_k - \frac{1}{L}\nabla f(y_k), \tag{0.8.2}$$

$$t_{k+1} = \frac{1 + \sqrt{1 + 4t_k^2}}{2}, \tag{0.8.3}$$

$$y_{k+1} = x_k + \frac{t_k - 1}{t_{k+1}}(x_k - x_{k-1}). \tag{0.8.4}$$

The scalar recursion is chosen so that

$$t_{k+1}^2 - t_{k+1} = t_k^2, \qquad t_k \geq \frac{k+1}{2}.$$

The proof begins with a one-step inequality. If $x^+ = y - L^{-1}\nabla f(y)$, then for every z,

$$f(x^+) \leq f(z) + \frac{L}{2}\left(\|z - y\|^2 - \left\|z - x^+\right\|^2\right). \tag{0.8.5}$$

Indeed, the descent lemma and convexity give

$$f(x^+) \leq f(y) + \langle \nabla f(y), x^+ - y\rangle + \frac{L}{2}\left\|x^+ - y\right\|^2,$$
$$f(y) \leq f(z) + \langle \nabla f(y), y - z\rangle.$$

Substituting $\nabla f(y) = L(y - x^+)$ and applying the three-point identity produces (0.8.5).

Theorem 0.8.1 (Accelerated rate)**.** *If f is convex and L-smooth, (0.8.2)–(0.8.4) satisfy*

$$f(x_k) - f(x^*) \le \frac{2L\,\|x_0 - x^*\|^2}{(k+1)^2}.$$

For the derivation, define

$$z_k = t_k x_k - (t_k - 1)x_{k-1}.$$

The extrapolation identity implies $t_{k+1}y_{k+1} - (t_{k+1} - 1)x_k = z_k$. Apply (0.8.5) at y_{k+1} once with $z = x_k$ and once with $z = x^*$, weight and add the inequalities, and use convexity together with $t_{k+1}^2 - t_{k+1} = t_k^2$. Completing squares yields the potential recursion

$$\frac{2t_{k+1}^2}{L}\big(f(x_{k+1}) - f^*\big) + \|z_{k+1} - x^*\|^2 \le \frac{2t_k^2}{L}\big(f(x_k) - f^*\big) + \|z_k - x^*\|^2. \tag{0.8.6}$$

The initial potential is at most $\|x_0 - x^*\|^2$. Dropping the nonnegative norm in (0.8.6) and using $t_k \ge (k+1)/2$ proves the rate. Notice that the potential decreases even though the objective values themselves need not be monotone.

0.8.3 Strongly convex acceleration

When f is also μ-strongly convex and μ is known, use constant extrapolation:

$$x_{k+1} = y_k - \frac{1}{L}\nabla f(y_k),$$
$$y_{k+1} = x_{k+1} + \beta(x_{k+1} - x_k), \qquad \beta = \frac{\sqrt{L} - \sqrt{\mu}}{\sqrt{L} + \sqrt{\mu}}.$$

Writing $q = \sqrt{\mu/L}$ and introducing $v_k = x_{k-1} + q^{-1}(x_k - x_{k-1})$, one can combine smoothness with the strong-convexity lower bound and complete squares to obtain the Lyapunov contraction

$$E_{k+1} \le (1-q)E_k, \qquad E_k = f(x_k) - f^* + \frac{\mu}{2}\|v_k - x^*\|^2.$$

Therefore

$$f(x_k) - f^* \le C\left(1 - \frac{1}{\sqrt{\kappa}}\right)^k.$$

Gradient descent needs $O(\kappa \log(1/\epsilon))$ iterations, whereas the accelerated complexity is $O(\sqrt{\kappa}\log(1/\epsilon))$.

0.8.4 FISTA

For the composite objective $F = f + h$, replace the gradient step by

$$x_k = \operatorname{prox}_{h/L}\left(y_k - \frac{1}{L}\nabla f(y_k)\right)$$

and keep the same t_k and extrapolation recursions. This is the fast iterative shrinkage-thresholding algorithm (FISTA). The proximal-gradient three-point estimate gives, for every z,

$$F(x^+) \le F(z) + \frac{L}{2}\left(\|z-y\|^2 - \left\|z-x^+\right\|^2\right).$$

Since this is exactly (0.8.5) with F replacing f, the same potential proof yields

$$F(x_k) - F^* \le \frac{2L\,\|x_0 - x^*\|^2}{(k+1)^2}.$$

Backtracking tests the quadratic upper model only for the smooth term f; no differentiability of h is needed.

0.8.5 Restart and safeguards

Momentum can become misaligned when the local curvature changes. Common safeguards include restarting with $t_k = 1$ when $F(x_k) > F(x_{k-1})$, or when

$$\langle x_k - x_{k-1}, x_{k+1} - x_k\rangle > 0.$$

Monotone FISTA chooses the better of the accelerated candidate and an ordinary proximal-gradient candidate. Restart often recovers nearly linear behavior on strongly convex problems even when μ is unknown.

0.9 Frank-Wolfe Method

Frank–Wolfe solves $\min_{x\in C} f(x)$ for a compact convex set C using a linear minimization oracle rather than projection:

$$\begin{aligned} s_k &\in \arg\min_{s\in C}\langle \nabla f(x_k), s\rangle, \\ d_k &= s_k - x_k, \\ x_{k+1} &= x_k + \gamma_k d_k, qquad \gamma_k \in [0,1]. \end{aligned}$$

The update is a convex combination of feasible points, so feasibility is automatic. When $C = \mathrm{conv}\{a_1,\dots,a_m\}$, each oracle call returns an atom, and x_k is a combination of at most $k+1$ atoms. This is useful when linear optimization is cheap but Euclidean projection is expensive, as on simplices, flow polytopes, and nuclear-norm balls.

0.9.1 The Frank–Wolfe gap

The Frank–Wolfe gap

$$g_{\mathrm{FW}}(x) = \max_{s\in C}\langle \nabla f(x), x - s\rangle = \langle \nabla f(x), x - s(x)\rangle$$

is both a stopping criterion and a computable certificate. Convexity gives

$$f(x) - f^* \le \langle \nabla f(x), x - x^* \rangle \le \max_{s \in C} \langle \nabla f(x), x - s \rangle = g_{\mathrm{FW}}(x).$$

The gap is nonnegative because $s = x$ is feasible in the maximum. Hence $g_{\mathrm{FW}}(x) = 0$ if and only if x is optimal.

0.9.2 Curvature and descent

An affine-invariant smoothness measure is the curvature constant

$$C_f = \sup_{\substack{x,s \in C \\ \gamma \in (0,1]}} \frac{2}{\gamma^2} \left[f(x + \gamma(s - x)) - f(x) - \gamma \langle \nabla f(x), s - x \rangle \right].$$

If ∇f is L-Lipschitz and C has diameter $D = \sup_{x,y \in C} \|x - y\|$, the descent lemma gives $C_f \le LD^2$. Using the oracle direction in the definition of curvature yields

$$f\Big(x + \gamma(s(x) - x)\Big) \le f(x) - \gamma g_{\mathrm{FW}}(x) + \frac{\gamma^2}{2} C_f. \tag{0.9.1}$$

Common choices are $\gamma_k = 2/(k+2)$, exact line search over $[0, 1]$, or the curvature-adaptive step

$$\gamma_k = \min \left\{ \frac{g_{\mathrm{FW}}(x_k)}{C_f}, 1 \right\}.$$

0.9.3 Primal convergence and gap complexity

Let $h_k = f(x_k) - f^*$. Since $h_k \le g_{\mathrm{FW}}(x_k)$, (0.9.1) gives the fundamental recursion

$$h_{k+1} \le (1 - \gamma_k) h_k + \frac{\gamma_k^2}{2} C_f. \tag{0.9.2}$$

Theorem 0.9.1 (Frank–Wolfe rate)**.** *With $\gamma_0 = 1$ and $\gamma_k = 2/(k+2)$ thereafter,*

$$f(x_k) - f^* \le \frac{2C_f}{k+2} leq \frac{2LD^2}{k+2}.$$

The same bound holds under exact line search.

For the proof, the first full step gives $h_1 \le C_f/2$. If $h_k \le 2C_f/(k+2)$, substitute $\gamma_k = 2/(k+2)$ into (0.9.2) to obtain

$$h_{k+1} \le \frac{2C_f(k+1)}{(k+2)^2} \le \frac{2C_f}{k+3},$$

where the last inequality follows from $(k+1)(k+3) \le (k+2)^2$. Induction proves the result.

The primal rate does not by itself bound the current Frank–Wolfe gap. With exact line search or minimization of the quadratic upper bound in (0.9.1), summing a constant test step over the window $k = K, \dots, 2K$ gives

$$\min_{K \le k \le 2K} g_{\mathrm{FW}}(x_k) \le \frac{4C_f}{K+2}.$$

Thus $O(C_f/\epsilon)$ oracle calls suffice to produce an explicit ϵ-optimality certificate.

0.9.4 Linear minimization oracles

For the probability simplex, the oracle places all mass on a smallest component of the gradient:

$$s = e_i, qquadi \in \arg\min_j [\nabla f(x)]_j.$$

For the ℓ_1 ball $\{x : \|x\|_1 \le \tau\}$,

$$s = -\tau \operatorname{sign}([\nabla f(x)]_i) e_i, qquadi \in \arg\max_j \left|[\nabla f(x)]_j\right|.$$

Each iteration introduces at most one new nonzero coordinate. For the nuclear-norm ball $\{X : \|X\|_* \le \tau\}$, if uv^T is a leading singular-vector pair of $\nabla f(X)$, then

$$S = -\tau uv^\mathsf{T}.$$

Only a leading singular-vector computation is required, and the iterate's rank increases by at most one.

0.9.5 Away steps and variants

Iterates are convex combinations of oracle outputs, often giving sparse or low-rank solutions early. Standard Frank–Wolfe can zig-zag near a boundary; it can add atoms quickly but remove their weights only gradually. If $x_k = \sum_{a \in \mathcal{A}_k} \lambda_a a$, an away step chooses

$$v_k \in \arg\max_{v \in \mathcal{A}_k} \langle \nabla f(x_k), v \rangle$$

and moves along $x_k - v_k$ up to the largest step preserving nonnegative coefficients. Pairwise Frank–Wolfe transfers weight directly from v_k to s_k. For strongly convex objectives over polytopes, these variants can converge linearly under a geometric-width condition. Fully corrective, approximate-oracle, stochastic, and block-coordinate variants are also useful when the active set or oracle has additional structure.

0.10 Proximal Point Algorithm

Let $F : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be proper, closed, and convex, with a nonempty solution set $X^* = \arg\min F$. The proximal point algorithm is

$$x_{k+1} = \operatorname{prox}_{\lambda_k F}(x_k) = \arg\min_x \left\{ F(x) + \frac{1}{2\lambda_k} \|x - x_k\|^2 \right\}. \tag{0.10.1}$$

Its optimality condition is

$$s_{k+1} := \frac{x_k - x_{k+1}}{\lambda_k} \in \partial F(x_{k+1}). \tag{0.10.2}$$

Thus PPA is an implicit subgradient method: unlike ordinary subgradient descent, it evaluates the subgradient at the new point.

Theorem 0.10.1 (Fejér monotonicity)**.** *If $x^* \in X^*$, every exact proximal point step satisfies*

$$\|x_{k+1} - x^*\|^2 \leq \|x_k - x^*\|^2 - \|x_{k+1} - x_k\|^2 .$$

If $\inf_k \lambda_k > 0$ and a minimizer exists, the iterates converge to a minimizer in finite dimensions.

To derive the inequality, use $0 \in \partial F(x^*)$ and monotonicity of the subdifferential with (0.10.2):

$$\langle x_{k+1} - x^*, x_k - x_{k+1} \rangle \geq 0.$$

Expanding

$$\|x_k - x^*\|^2 = \|x_k - x_{k+1}\|^2 + \|x_{k+1} - x^*\|^2 + 2\langle x_k - x_{k+1}, x_{k+1} - x^* \rangle$$

proves Fejér decrease. Hence the sequence is bounded and $\sum_k \|x_{k+1} - x_k\|^2 < \infty$. If $\inf_k \lambda_k > 0$, then $s_{k+1} \to 0$; closedness of the subdifferential graph makes every cluster point optimal, and Fejér monotonicity forces the whole sequence to converge.

For a constant parameter $\lambda_k \equiv \lambda$, the subgradient inequality and the same three-point identity give

$$\begin{aligned} F(x_{j+1}) - F^* &\leq \langle s_{j+1}, x_{j+1} - x^* \rangle \\ &= \frac{1}{2\lambda} \left(\|x_j - x^*\|^2 - \|x_{j+1} - x^*\|^2 - \|x_{j+1} - x_j\|^2 \right). \end{aligned}$$

Summing telescopes. Since each proximal step decreases F,

$$F(x_k) - F^* \leq \frac{\|x_0 - x^*\|^2}{2\lambda k}.$$

0.10.1 Resolvents and the Moreau envelope

For a set-valued operator T, its resolvent is $J_{\lambda T} = (I + \lambda T)^{-1}$. Condition (0.10.2) says

$$\operatorname{prox}_{\lambda F} = J_{\lambda \partial F}.$$

This extends PPA to finding a zero of a maximal monotone operator. The Fejér proof uses only monotonicity and therefore carries over unchanged.

The Moreau envelope

$$F_\lambda(x) = \min_z \left\{ F(z) + \frac{1}{2\lambda} \|z - x\|^2 \right\}$$

is convex and continuously differentiable, with

$$\nabla F_\lambda(x) = \frac{1}{\lambda} \left(x - \operatorname{prox}_{\lambda F}(x) \right). \tag{0.10.3}$$

Danskin's theorem gives the formula. Firm nonexpansiveness of the proximal map makes $I - \operatorname{prox}_{\lambda F}$ firmly nonexpansive, so ∇F_λ is $1/\lambda$-Lipschitz. Consequently, PPA is gradient descent with step λ_k on the smooth surrogate F_{λ_k}.

0.10.2 Inexact proximal steps

Exact solution of the proximal subproblem may be as hard as the original problem. A standard inexact condition is

$$\frac{x_k - x_{k+1}}{\lambda_k} - e_{k+1} \in \partial F(x_{k+1}), \qquad \sum_{k=0}^{\infty} \lambda_k \|e_{k+1}\| < \infty.$$

The summability requirement makes the accumulated perturbation finite, and a quasi-Fejér version of the preceding proof gives convergence. Relative-error criteria scale the inner tolerance with $\|x_{k+1} - x_k\|$ and avoid oversolving early subproblems.

PPA is robust but each step may be as difficult as the original problem. Its importance is conceptual: proximal gradient, augmented Lagrangian, and ADMM can be interpreted as tractable approximations or operator-splitting forms of proximal point iterations.

0.11 Augmented Lagrangian Method

For equality constraints

$$\min_x f(x) \quad \text{subject to} \quad Ax = b,$$

the augmented Lagrangian is

$$\mathcal{L}_\rho(x, y) = f(x) + y^\mathsf{T}(Ax - b) + \frac{\rho}{2}\|Ax - b\|^2. \tag{0.11.1}$$

Allowing a possibly varying penalty $\rho_k > 0$, the method of multipliers alternates

$$x_{k+1} \in \arg\min_x \mathcal{L}_{\rho_k}(x, y_k), \qquad y_{k+1} = y_k + \rho_k(Ax_{k+1} - b).$$

The quadratic penalty improves feasibility without requiring $\rho_k \to \infty$, while the dual variable accumulates constraint violations.

Under convexity and standard constraint qualifications, a pair (x^*, y^*) is optimal when

$$0 \in \partial f(x^*) + A^\mathsf{T} y^*, \qquad Ax^* = b.$$

These KKT residuals should be monitored. Inner minimization need not be exact, but its tolerances must become sufficiently accurate for convergence theory.

0.11.1 ALM as proximal point on the dual

Let

$$d(y) = \inf_x\{f(x) + \langle y, Ax - b\rangle\}, \qquad \phi(y) = -d(y).$$

Under standard regularity, ϕ is closed and convex. Optimality of the x-subproblem gives

$$\begin{aligned} 0 &\in \partial f(x_{k+1}) + A^\mathsf{T}\Big(y_k + \rho_k(Ax_{k+1} - b)\Big) \\ &= \partial f(x_{k+1}) + A^\mathsf{T} y_{k+1}. \end{aligned}$$

Thus x_{k+1} minimizes the ordinary Lagrangian at y_{k+1}. The dual subgradient formula then gives

$$Ax_{k+1} - b \in \partial d(y_{k+1}), \qquad b - Ax_{k+1} \in \partial \phi(y_{k+1}).$$

Using the multiplier update,

$$0 \in \partial \phi(y_{k+1}) + \frac{1}{\rho_k}(y_{k+1} - y_k),$$

which is exactly the proximal optimality condition. Therefore

$$y_{k+1} = \operatorname{prox}_{\rho_k \phi}(y_k) = \arg\min_y \left\{ -d(y) + \frac{1}{2\rho_k} \|y - y_k\|^2 \right\}. \tag{0.11.2}$$

If the dual solution set is nonempty, applying PPA's Fejér inequality to (0.11.2) yields, for every dual solution y^*,

$$\|y_{k+1} - y^*\|^2 \le \|y_k - y^*\|^2 - \rho_k^2 \|Ax_{k+1} - b\|^2 .$$

Hence, when $\inf_k \rho_k > 0$, the multipliers converge to a dual solution and $Ax_{k+1} - b \to 0$. The same summable-error principle used for inexact PPA allows approximate solution of the ALM subproblems without requiring them to be solved excessively accurately at early iterations.

0.12 Alternating Direction Method of Multipliers

ADMM addresses

$$\min_{x,z} f(x) + h(z) \quad \text{subject to} \quad Ax + Bz = c,$$

where both functions may be nonsmooth. Its augmented Lagrangian is

$$\mathcal{L}_\rho(x, z, y) = f(x) + h(z) + \langle y, Ax + Bz - c \rangle + \frac{\rho}{2} \|Ax + Bz - c\|^2 .$$

Whereas ALM jointly minimizes over (x, z), ADMM performs one Gauss–Seidel sweep followed by the multiplier update:

$$\begin{aligned} x_{k+1} &\in \arg\min_x \mathcal{L}_\rho(x, z_k, y_k), \\ z_{k+1} &\in \arg\min_z \mathcal{L}_\rho(x_{k+1}, z, y_k), \\ y_{k+1} &= y_k + \rho(Ax_{k+1} + Bz_{k+1} - c). \end{aligned}$$

Using the scaled dual variable $u = y/\rho$, the iterations are

$$\begin{aligned} x_{k+1} &= \arg\min_x \left\{ f(x) + \frac{\rho}{2} \|Ax + Bz_k - c + u_k\|^2 \right\}, \\ z_{k+1} &= \arg\min_z \left\{ g(z) + \frac{\rho}{2} \|Ax_{k+1} + Bz - c + u_k\|^2 \right\}, \\ u_{k+1} &= u_k + Ax_{k+1} + Bz_{k+1} - c. \end{aligned}$$

Separating x and z can turn a difficult joint problem into two simple subproblems.

0.12.1 Residuals and stopping criteria

Define the primal and dual residuals by

$$r_{k+1} = Ax_{k+1} + Bz_{k+1} - c, \qquad s_{k+1} = \rho A^{\mathsf{T}} B(z_{k+1} - z_k). \tag{0.12.1}$$

To see the second formula, optimality of the z update and the multiplier update give

$$0 \in \partial h(z_{k+1}) + B^{\mathsf{T}} y_{k+1}.$$

The x-optimality condition becomes

$$0 \in \partial f(x_{k+1}) + A^{\mathsf{T}} y_{k+1} - \rho A^{\mathsf{T}} B(z_{k+1} - z_k).$$

Thus a small primal residual enforces feasibility, while a small dual residual makes x-stationarity accurate. For the constraint $x = z$, the norm of the dual residual is simply $\rho \|z_{k+1} - z_k\|$.

Stop only when both residuals are small relative to absolute and relative tolerances. One scale-aware choice is

$$\begin{aligned} \epsilon_{\text{pri}} &= \sqrt{p}\,\epsilon_{\text{abs}} + \epsilon_{\text{rel}} \max\{\|Ax_k\|, \|Bz_k\|, \|c\|\}, \\ \epsilon_{\text{dual}} &= \sqrt{n}\,\epsilon_{\text{abs}} + \epsilon_{\text{rel}} \left\|A^{\mathsf{T}} y_k\right\|. \end{aligned}$$

0.12.2 Global convergence

Assume f and h are proper, closed, and convex, a saddle point (x^*, z^*, y^*) exists, and both subproblems are solved exactly. Define

$$V_k = \frac{1}{\rho} \|y_k - y^*\|^2 + \rho \|B(z_k - z^*)\|^2.$$

Theorem 0.12.1 (ADMM Lyapunov decrease)**.** *For fixed $\rho > 0$,*

$$V_{k+1} \le V_k - \rho \|r_{k+1}\|^2 - \rho \|B(z_{k+1} - z_k)\|^2. \tag{0.12.2}$$

Consequently, $r_k \to 0$, $s_k \to 0$, and objective values converge to the optimum. If A and B have full column rank, the entire primal–dual sequence converges to a solution.

Here is the main algebra behind the result. The x-stationarity relation and the KKT condition at (x^*, z^*, y^*) give, by monotonicity of ∂f,

$$-\langle A(x_{k+1} - x^*), y_{k+1} - y^* \rangle + \rho \langle A(x_{k+1} - x^*), B(z_{k+1} - z_k) \rangle \ge 0.$$

Likewise, monotonicity of ∂h gives

$$-\langle B(z_{k+1} - z^*), y_{k+1} - y^* \rangle \ge 0.$$

Add the inequalities, use primal feasibility of (x^*, z^*), the identity $y_{k+1} - y_k = \rho r_{k+1}$, and monotonicity between consecutive z updates. Applying

$$2\langle a-b, a-c\rangle = \|a-c\|^2 - \|b-c\|^2 + \|a-b\|^2$$

to the dual and Bz terms yields (0.12.2). Summing it proves residual convergence. Subgradient inequalities give the objective sandwich

$$-\langle y^*, r_k\rangle \le f(x_k) + h(z_k) - p^* \le -\langle y_k, r_k\rangle + \rho\langle B(z_k - z_{k-1}), r_k - B(z_k - z^*)\rangle,$$

whose two sides tend to zero by the Lyapunov bound.

0.12.3 Two-function splitting and Lasso

To minimize $f(x) + h(x)$, introduce a copy z and impose $x - z = 0$. Then

$$\begin{aligned} x_{k+1} &= \operatorname{prox}_{f/\rho}(z_k - u_k), \\ z_{k+1} &= \operatorname{prox}_{h/\rho}(x_{k+1} + u_k), \\ u_{k+1} &= u_k + x_{k+1} - z_{k+1}. \end{aligned}$$

This is useful when the individual proximal maps are simpler than the proximal map of their sum.

For the Lasso

$$\min_x \frac{1}{2}\|Dx - b\|^2 + \lambda \|x\|_1,$$

the split $x = z$ gives

$$\begin{aligned} (D^\mathsf{T}D + \rho I)x_{k+1} &= D^\mathsf{T}b + \rho(z_k - u_k), \\ z_{k+1} &= \operatorname{soft}_{\lambda/\rho}(x_{k+1} + u_k), \\ u_{k+1} &= u_k + x_{k+1} - z_{k+1}. \end{aligned}$$

For fixed ρ, factor $D^\mathsf{T}D + \rho I$ once and reuse triangular solves rather than forming an inverse.

Matrix problems admit the same pattern. Splitting a nuclear norm from a simple constraint produces a singular-value-thresholding update followed by a projection; splitting entrywise sparsity from a PSD constraint alternates soft thresholding with eigenvalue clipping.

0.12.4 Consensus and penalty adaptation

For distributed learning with local objectives f_i, rewrite

$$\min_z \sum_{i=1}^m f_i(z) \quad \text{as} \quad \min_{x_1,\dots,x_m,z} \sum_{i=1}^m f_i(x_i) \quad \text{subject to } x_i = z.$$

ADMM separates the local x_i updates, while the z update averages the shifted local variables; only that aggregate must be communicated.

A large ρ emphasizes primal feasibility but can slow dual progress, and a small ρ does the reverse. Residual balancing increases ρ when $\|r\|$ dominates $\|s\|$ and decreases it in the opposite case. Because $u = y/\rho$, changing ρ requires rescaling u to preserve the same unscaled multiplier. Over-relaxation replaces Ax_{k+1} in the z and dual updates by a weighted combination of Ax_{k+1} and the previous affine term; values moderately above one often improve practical convergence.

0.13 Coordinate Descent Method

Coordinate descent updates one block at a time:

$$x_j^{k+1} \in \arg\min_t f(x_1^{k+1}, \dots, x_{j-1}^{k+1}, t, x_{j+1}^k, \dots, x_d^k).$$

Updates may be cyclic, randomized, greedy, or performed in blocks. For a smooth objective with coordinate Lipschitz constant L_j, a coordinate gradient step is $x_j \leftarrow x_j - L_j^{-1}\partial_j f(x)$.

Theorem 0.13.1 (Randomized coordinate rate)**.** *For convex smooth f with coordinate-wise Lipschitz gradients, randomized coordinate descent has an expected $O(1/k)$ objective gap. Under strong convexity it converges linearly in expectation, with a rate controlled by the coordinate smoothness constants.*

Coordinate methods are effective when updates are closed form and data are sparse. For lasso, each update is soft thresholding. Cache residuals rather than recomputing $X\beta$ after each coordinate. Pure coordinate minimization can fail for coupled nonsmooth nonconvex objectives, so assumptions matter.

0.14 Stochastic Gradient Descent

For finite-sum or expectation objectives,

$$f(x) = \frac{1}{n}\sum_{i=1}^n f_i(x) \quad \text{or} \quad f(x) = \mathbb{E}_\xi[F(x;\xi)],$$

stochastic gradient descent (SGD) uses

$$x_{k+1} = x_k - \alpha_k g_k, \qquad \mathbb{E}[g_k \mid x_k] = \nabla f(x_k). \tag{0.14.1}$$

A minibatch average lowers gradient variance roughly inversely with batch size until correlations and hardware effects dominate.

Theorem 0.14.1 (Strongly convex SGD rate)**.** *Suppose f is μ-strongly convex, stochastic gradients are unbiased with bounded second moment, and α_k is of order $1/k$. Then*

$$\mathbb{E}[f(\bar{x}_k) - f(x^*)] = O(k^{-1})$$

for a suitable averaged iterate. For convex but not strongly convex objectives, the standard rate is $O(k^{-1/2})$.

A constant learning rate reaches a noise neighborhood rather than converging exactly, but is useful for rapid initial progress. Decay reduces the steady-state variance. Momentum updates

$$v_{k+1} = \beta v_k + g_k, \qquad x_{k+1} = x_k - \alpha v_{k+1}$$

smooth noisy directions and accelerate persistent ones.

Shuffle finite datasets between epochs. Report gradient evaluations or data passes, not iterations alone, when comparing batch sizes. For nonconvex smooth objectives, a common result controls the average squared gradient norm, so convergence means approaching stationarity rather than a global minimum.

0.15 Adaptive Methods

Adaptive methods rescale coordinates using past gradients:

$$x_{k+1} = x_k - \alpha D_k^{-1} m_k.$$

They are helpful for sparse gradients and poorly scaled coordinates. However, their implicit geometry, memory cost, and generalization can differ from SGD. The stabilizer ε is part of the algorithm, not merely a guard against division by zero.

0.15.1 ADAM

Adam maintains exponential first and second moments:

$$\begin{aligned} m_k &= \beta_1 m_{k-1} + (1-\beta_1) g_k, \\ v_k &= \beta_2 v_{k-1} + (1-\beta_2) g_k^{\odot 2}, \\ \widehat{m}_k &= m_k/(1-\beta_1^k), \qquad \widehat{v}_k = v_k/(1-\beta_2^k), \\ x_{k+1} &= x_k - \alpha \frac{\widehat{m}_k}{\sqrt{\widehat{v}_k} + \varepsilon}. \end{aligned}$$

Bias correction matters early because moments start at zero.

Vanilla Adam does not converge for every convex online problem; AMSGrad repairs a problematic decrease in the effective second-moment accumulator by using a coordinatewise running maximum. In deep learning, AdamW decouples weight decay from the adaptive gradient update. Adding λx to the gradient is not equivalent to multiplicative weight decay under a coordinate-dependent preconditioner.

0.15.2 AdaGrad

Diagonal AdaGrad uses

$$v_k = v_{k-1} + g_k^{\odot 2}, \qquad x_{k+1} = x_k - \alpha \frac{g_k}{\sqrt{v_k} + \varepsilon}.$$

Coordinates with frequent large gradients receive smaller future steps.

Theorem 0.15.1 (AdaGrad regret, informal)**.** *For convex online losses and a bounded domain, diagonal AdaGrad has regret controlled by*

$$\sum_{j=1}^{d} \sqrt{\sum_{k=1}^{T} g_{k,j}^2},$$

up to a factor depending on the domain diameter. This can be much smaller than a bound based on the worst gradient when gradients are sparse.

Because v_k only increases, learning rates may decay too aggressively in a long nonstationary run. RMSProp and Adam use exponential rather than cumulative second moments.

0.16 Shampoo and Muon

Diagonal adaptive methods ignore correlation between parameter coordinates. Full-matrix AdaGrad captures it but requires prohibitive storage and matrix roots. Shampoo exploits tensor structure. For a matrix gradient $G_k \in \mathbb{R}^{m\times n}$, a basic form accumulates

$$L_k = \varepsilon I_m + \sum_{t\le k} G_t G_t^{\mathsf{T}}, \qquad R_k = \varepsilon I_n + \sum_{t\le k} G_t^{\mathsf{T}} G_t$$

and preconditions by

$$\widetilde{G}_k = L_k^{-1/4} G_k R_k^{-1/4}. \tag{0.16.1}$$

Higher-order tensors receive one preconditioner per mode. Matrix roots are expensive, so practical implementations update them intermittently, use blocking, and combine them with grafting or momentum. Damping controls nearly singular eigendirections.

Muon is designed primarily for matrix-valued hidden-layer weights. It forms a momentum matrix M_k and approximately replaces its singular values by one,

$$M_k = U\Sigma V^{\mathsf{T}} \quad \longmapsto \quad \operatorname{polar}(M_k) = UV^{\mathsf{T}},$$

usually through a small number of Newton–Schulz matrix iterations rather than an SVD. The resulting direction is then scaled and used for the parameter update. Vector parameters and some embedding or output parameters are commonly handled by another optimizer.

Shampoo uses accumulated second-moment geometry; Muon orthogonalizes a current momentum direction. They are therefore not interchangeable despite both using matrix operations. Exact scaling conventions vary by implementation, so a reproducible description must specify normalization, Newton–Schulz polynomial, iteration count, momentum, weight decay, and parameter groups.

0.17 Automatic Differentiation

Automatic differentiation (AD) applies the chain rule to a computational graph. It is neither symbolic differentiation nor finite differences. For $f : \mathbb{R}^n \to \mathbb{R}^m$ with Jacobian $J_f(x)$:

- forward mode computes a Jacobian–vector product $J_f(x)v$;
- reverse mode computes a vector–Jacobian product $u^{\mathsf{T}} J_f(x)$.

Forward mode is attractive when inputs are few; reverse mode is attractive for a scalar loss with many parameters, because one reverse sweep obtains the full gradient.

Example 0.17.1 (Reverse-mode chain rule)**.** If $z = Wx$, $a = \tanh z$, and $\ell = \frac{1}{2}\|a - y\|^2$, reverse mode computes

$$\bar{a} = a - y, \qquad \bar{z} = \bar{a} \odot (1 - \tanh^2 z), \qquad \nabla_W \ell = \bar{z} x^{\mathsf{T}}.$$

The barred variables are adjoints: derivatives of the final loss with respect to intermediate values.

Reverse mode stores intermediate activations, trading memory for computation. Checkpointing recomputes selected activations during the backward pass. Higher-order derivatives are obtained by composing modes; Hessian–vector products can be computed without forming the Hessian.

Common bugs arise from in-place mutation, unintended detachment of the graph, nondifferentiable control flow, and unstable primitives. Gradient checking with centered finite differences is useful on small problems, but step sizes that are too small suffer cancellation and those too large suffer truncation error.

0.18 Applications

Optimization algorithms should be chosen from problem structure: smoothness, convexity, separability, sparsity, constraint geometry, and data-access cost. The following examples connect the general methods to recurring machine learning and inverse-problem formulations.

0.18.1 Matrix Factorization

Given a partially observed matrix M, low-rank factorization solves

$$\min_{U \in \mathbb{R}^{m \times r},\, V \in \mathbb{R}^{n \times r}} \frac{1}{2} \sum_{(i,j) \in \Omega} (M_{ij} - u_i^{\mathsf{T}} v_j)^2 + \frac{\lambda}{2}(\|U\|_F^2 + \|V\|_F^2). \tag{0.18.1}$$

The problem is nonconvex jointly but convex in either factor with the other fixed. Alternating least squares exploits this biconvex structure. SGD samples an observed entry and updates only u_i and v_j:

$$e_{ij} = M_{ij} - u_i^{\mathsf{T}} v_j, \quad u_i \leftarrow u_i + \alpha(e_{ij} v_j - \lambda u_i), \quad v_j \leftarrow v_j + \alpha(e_{ij} u_i - \lambda v_j),$$

where the second update should use either the old u_i or a documented sequential convention.

The factors are nonidentifiable: $UV^{\mathsf{T}} = (UA)(VA^{-\mathsf{T}})^{\mathsf{T}}$ for invertible A. Prediction of the product may be well defined even when individual factors are not. Random initialization, regularization, and validation of rank r are essential.

The convex alternative minimizes a data-fit term plus nuclear norm $\|X\|_* = \sum_j \sigma_j(X)$. Its proximal operator performs singular-value soft thresholding, while Frank–Wolfe on a nuclear-norm ball adds rank-one directions using leading singular vectors.

0.18.2 Support Vector Machines

The linear soft-margin SVM has a smooth quadratic regularizer and nonsmooth hinge losses:

$$\min_{\beta_0,\beta} \frac{\lambda}{2}\|\beta\|^2 + \frac{1}{n}\sum_{i=1}^{n}[1 - y_i(\beta_0 + x_i^\mathsf{T}\beta)]_+.$$

Subgradient methods are simple but slow. Coordinate descent is effective in the dual; stochastic subgradient descent scales to large sparse primal problems. For one observation, a subgradient with respect to β is

$$\lambda\beta - y_i x_i \mathbb{1}\{y_i(\beta_0 + x_i^\mathsf{T}\beta) < 1\},$$

with a set-valued choice exactly at margin one.

Squared hinge loss is differentiable once but changes the robustness and dual. Kernel methods favor dual solvers, while high-dimensional sparse linear SVMs usually favor primal methods. Optimization tolerance should be judged relative to statistical error: solving the training objective far beyond the precision supported by the data wastes computation.

0.18.3 Compressed Sensing

Compressed sensing recovers a sparse vector $x_0 \in \mathbb{R}^d$ from $b = Ax_0 + \varepsilon$ with fewer measurements than ambient coordinates. The ideal problem

$$\min_x \|x\|_0 \quad \text{subject to} \quad Ax = b$$

is combinatorial. Basis pursuit replaces cardinality by the convex ℓ_1 norm:

$$\min_x \|x\|_1 \quad \text{subject to} \quad Ax = b. \tag{0.18.2}$$

With noise, common forms are

$$\min_x \frac{1}{2}\|Ax - b\|^2 + \lambda\|x\|_1 \quad \text{or} \quad \min_x \|x\|_1 \text{ subject to } \|Ax - b\| \le \delta.$$

ISTA, FISTA, ADMM, and coordinate descent apply directly.

Definition 0.18.1 (Restricted isometry property)**.** A matrix A has restricted isometry constant δ_s if δ_s is the smallest number such that

$$(1 - \delta_s)\|x\|^2 \le \|Ax\|^2 \le (1 + \delta_s)\|x\|^2$$

for every s-sparse x.

Theorem 0.18.1 (Sparse recovery, representative form)**.** *If A satisfies a sufficiently strong restricted isometry condition, such as $\delta_{2s} < \sqrt{2} - 1$, then basis pursuit exactly recovers every s-sparse vector from noiseless measurements. With noise and approximate sparsity, the reconstruction error is bounded by a noise term plus the best s-term approximation error.*

Random sub-Gaussian measurement matrices satisfy suitable RIP conditions with roughly $s \log(d/s)$ measurements. Feature standardization and the choice of λ remain important in practical statistical problems.

0.18.4 Deep Learning

A neural network is a composition

$$h_0 = x, \qquad z_\ell = W_\ell h_{\ell-1} + b_\ell, \qquad h_\ell = \sigma_\ell(z_\ell),$$

trained by minimizing an empirical loss, usually with regularization. Reverse- mode AD is backpropagation applied to this graph.

The objective is nonconvex and has symmetries: hidden units may be permuted, and homogeneous networks allow compensating rescalings across layers. Consequently parameter identifiability and Hessian positive definiteness are not expected. Practical optimization seeks a useful low-loss stationary region, not a certified global minimum.

Initialization controls signal and gradient propagation. Xavier scaling is suited to approximately symmetric activations; He scaling is suited to ReLU- like activations. Normalization, residual connections, and careful parameter scaling improve conditioning. Gradient clipping controls rare large updates but changes the optimization direction.

Minibatch SGD and Adam-type methods dominate because full gradients are expensive. A typical training loop separates:

1. forward evaluation and loss computation;
2. reverse-mode gradient computation;
3. optional clipping or gradient accumulation;
4. optimizer and learning-rate-schedule updates;
5. validation under evaluation-mode behavior.

Weight decay, data augmentation, dropout, and early stopping are regularization choices as well as optimization choices. Training loss alone cannot choose them. Reproducible computation records random seeds, data order, precision, optimizer conventions, schedules, and checkpoint selection.

Mixed-precision training increases throughput but can underflow small gradients; loss scaling and higher-precision accumulators address this. Distributed data-parallel training averages gradients across workers. Very large batches reduce gradient noise but often require learning-rate and warmup adjustments.

Remark 0.18.1. For both classical and deep models, optimization error, estimation error, and approximation error are distinct. A lower training objective does not imply a lower population risk, and an optimizer comparison is meaningful only when data processing, model, tuning budget, and stopping rules are controlled.